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*Environmental Psychology (Psy511) Finalterm Highlighted Handouts
by Be kind to yourself.*



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Lecture #19

Impact of Environment on Individual (Personality Development and Individual Differences)

Personality:

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"Personality consists of all the relatively stable and distinctive styles of thought, behavior and emotional response that characterize a person's adaptations to surrounding circumstances." (Maddi 1976, Mischel 1976).

Personality can also be defined as a dynamic and organized set of characteristics possessed by a person that uniquely influences his or her cognitions, motivations, and behaviors in various situations. The word "personality" originates from the Latin word *persona*, which means mask. For example, someone might be described as outgoing, empathetic, and shy, indicating traits that make up their personality.

Personality Development:

Emergence of distinctive style of thoughts, feelings, and behaviour that make each human being a unique individual.

Two questions for Personality Psychologists:

1. Individual Differences

When several people encounter the same situation, why don't they all react alike?

2. Environmental impact

What is the relationship between human behavior and environment?

Personality Research – Theoretical Frameworks

- Trait Theory
- Murray's Theory of Personality
- Interactionist Perspective
- Metron's Notion of Self-Fulfilling Prophecy

Trait Theories of Personality Trait:

Relatively enduring way in which an individual differs from others.

- Personality traits are relatively stable over time.
- Personality traits are consistent over situations.
- Intensity of traits.

1. Gordon Allport Trait Theory:

Allport & Odbert developed the first trait theory, and they categorized traits into three levels (1961).

Cardinal traits:

These traits dominate and shape a person's behavior, so much so that they seem to become synonymous with the person. These include the need for money, ambition, etc.

Central traits:

These traits are less dominating, and they are present in all human beings to different degrees.

For example, "honesty" or "intelligence".

Secondary traits:

These traits are very specific behaviors or preferences that appear in certain contexts, say being impatient while waiting.

2. Raymond Cattell

Cattell reduced the size of Allport's list to make it more manageable and created a science of personality. In this way, he found out that personality consists of 16 distinct units, which he called source traits (1978).

3. Hans Eysenck

Hans Eysenck felt that even five factors were too many. So, he created a three-dimension model, which included extraversion, neuroticism, and psychoticism. Eysenck also claimed that individual differences are biologically rooted: introverts have central nervous systems that are more sensitive to stimulation and so avoid intense sources of excitement. (1967)

Lesson # 20

Murray's Theory of Personality Development And Interactionist Perspective

Henry Murray and Psychogenic Needs

American psychologist Henry Murray (1893-1988) developed a theory of personality that was organized in terms of motives, presses, and needs. Murray described a needs as a, "potentiality or readiness to respond in a certain way under certain given circumstances" (1938)

Theories of personality based upon needs and motives suggest that our personalities are a reflection of behaviors controlled by needs. While some needs are temporary and changing, other needs are more deeply seated in our nature. According to Murray, these psychogenic needs function mostly on the unconscious level, but play a major role in our personality.

Murray's Types of Needs

Murray identified needs of two types:

1. Primary Needs

Primary needs are based upon biological demands, such as the need for oxygen, food, and water.

2. Secondary Needs

Secondary needs are generally psychological, such as the need for nurturing, independence, and achievement.

List of Psychogenic Needs

The following is a partial list of 24 needs identified by Murray and his colleagues. According to Murray, all people have these needs, but each individual tends to have a certain level of each need.

1. Ambition Needs

- Achievement: Success, accomplishment, and overcoming obstacles.
- Exhibition: Shocking or thrilling other people.
- Recognition: Displaying achievements and gaining social status.

2. Materialistic Needs

- Acquisition: Obtaining things.
- Construction: Creating things.
- Order: Making things neat and organized.
- Retention: Keeping things.

3. Power Needs

- Abasement: Confessing and apologizing.
- Autonomy: Independence and resistance.
- Aggression: Attacking or ridiculing others.
- Blame Avoidance: Following the rules and avoiding blame.
- Deference: Obeying and cooperating with others.

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- Dominance: Controlling others.

4. Affection Needs

- Affiliation: Spending time with other people.
- Nurturance: Taking care of another person.
- Play: Having fun with others.
- Rejection: Rejecting other people.
- Succorance: Being helped or protected by others.

5. Information Needs

- Cognizance: Seeking knowledge and asking questions.
- Exposition: Education others.

Interactionism: The interactionist perspective on the situation vs. person

Traits and Situations interact to influence behavior - how else could it be? (Its like the genetics vs. environment issue, one cannot exist without the other). So, the trait and situationist perspectives are too simplistic: reality is more complex. In reality, different situations affect different people in different ways. Some situations allow expression of personality; other situations provoke a narrower range of behavior. Thus, Behavior = personality x interpretation of the situation. It is vital to appreciate that there are individual differences in the personality-situation relationship.

Research (Kenrick et al, 1990) has shown that a trait will show up only in a situation where it is relevant. So anxiety may show up as a predictor of behavior some situations, and not others. Also, some situations allow expression of personality; others provoke narrower range of behavior.

Mischel's Contention (1968)

Human behaviour across situations is unpredictable in terms of personal factors alone.

Walter Mischel's contention revolves around the interaction between individuals and their environments. He suggested that behavior is not solely a result of internal traits but is significantly influenced by the external situation and context. Mischel argued that an individual's behavior is better predicted by understanding the specific circumstances they are in, rather than relying on broad trait-based predictions.

Mischel's view challenges traditional trait theories that suggest personality traits are consistent across different situations. Instead, he posited that there is a dynamic interplay between the person and the situation, where the environment can shape behavior just as much as an individual's personality traits can.

It emphasizes the role of situational factors in shaping behavior and suggests that modifying the environment can lead to changes in behavior. Mischel's work encourages a more integrative approach that considers both personal characteristics and environmental influence.

Behavior Contingency Units

Behavior Contingency Units are the close association between a person's behavior and the stimulus or cue properties of the setting in which it occurs. Substantial proportion of behavioral variance is accounted for by situational variables.

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Bowers, Bem and Allen's Viewpoint

- Situations are as much a product of the person as the person's behavior is function of the situation.
- The individual's capacity to shape the environment is evidenced quite clearly through behavioral processes such as art, architecture and community planning.
- Cognitive processes also play significant role in structuring the environment.

Metron's Self Fulfilling Prophecy Self Fulfilling Prophecy

Person's expectations about other people that lead him to act in a way that brings out the traits that he expects them to have. This is where interactionsit view of the environment and behavior assumes a dynamic interchange between man and the environment in which people affect, and are affected by their settings.

Lesson # 21

ENVIRONMENTAL CHANGES AND STRESS

Humans are incredibly adaptable; when not satisfied with their lot, they have the intelligence and ingenuity to create new things, to adapt to what is available, and even to adjust or alter their living environment to make it more congenial. But this flexibility is daily being challenged. Forces from within the species (e.g., violent crimes, war, acts of terrorism, and genocide), widespread natural catastrophes (e.g., famines, floods, droughts, earthquakes, and volcanic eruptions), and ever increasing and dangerous technological developments (e.g., faster automobiles, increasing numbers of aircraft attempting to occupy the same air corridors, escalating numbers of chemicals, and the proliferation of nuclear devices) are coalescing to test the limits of human adaptability. Additionally, interpersonal forces demanding more material goods and greater and more efficient provision of services are increasingly straining limited physical and human resources.

Thus, over and over again in the course of daily living, we witness threats to human adaptability, feel pressure for increased ingenuity to provide protection from external forces, and struggle to reconcile material possessions with individual desires. The operation of these forces is inextricably tied to what has commonly come to be labelled **stress**.

Technological Advancements:

The rapid pace of technological progress, while offering numerous benefits, also introduces new risks. The increase in high-speed transportation options, such as faster automobiles and densely populated air corridors, has led to heightened potential for accidents and collisions. The chemical industry's expansion has resulted in a vast array of substances entering the environment, many with unknown long-term effects on human health and ecosystems. Furthermore, the spread of nuclear technology raises concerns about safety, waste management, and the possibility of catastrophic events.

But what exactly is stress?

Personal accounts of stress, scales for measuring it, instructions for coping with it, and personality tests to see how well one can endure it have found their way into print and are supplemented by TV and radio accounts featuring self-proclaimed "experts" in its early detection and management.

Defining Stress: Theoretical Perspectives

Stress has been defined as a state that occurs when people are faced with demands from the environment that require them to change in some way. Most researchers agree with this definition. What they do not agree on is whether stress is the demand itself or the person's response to that demand.

Theoretical Models of Stress

Several models have been proposed to explain the stress phenomenon:

- **The James-Lange Theory of Emotion:** Suggests that emotions arise from physiological responses to events. According to this theory, stress would be considered the physical response to a demand.
- **The Cannon-Bard Theory:** Posits that emotions and physiological responses occur simultaneously but independently. Here, stress could be seen as both the demand and the response.
- **The Schachter-Singer Theory:** Asserts that emotions depend on physiological arousal and the cognitive interpretation of that arousal. In this view, stress involves an element of personal interpretation or appraisal of the demand.

Response Based Definition of Stress

The response-based definition of stress emphasizes the physiological & psychological reactions that occur when an individual is faced with demanding or threatening situations.

- **Physiological Responses:**

Response of the body to any demand, whether it is caused by, or results in, pleasant or unpleasant conditions." According to this definition, stress is conceptualized chiefly in terms of the body's physiological reaction to any demand placed on it, like galvanic skin response, change in heart rate and blood pressure etc.

Key Physiological Changes During Stress

- **Cardiovascular System:** During stress, heart rate and blood pressure increase to supply more oxygen and nutrients to the muscles, preparing the body for action.
- **Respiratory System:** Breathing becomes more rapid to distribute oxygen quickly throughout the body.
- **Musculoskeletal System:** Muscles tense up, readying the body for physical activity.
- **Endocrine System:** Stress hormones like adrenaline and cortisol are released, triggering the body's alert systems

- **Psychological Responses:**

The psychological response to stress typically involves a range of emotional and cognitive reactions. These can include feelings of anxiety, worry, or fear, as well as difficulty concentrating, confusion, loss of individual control, and loss of self esteem.

Emotional Reactions Under Stress

- **Anxiety:** A heightened state of alertness that prepares an individual to face potential threats.
- **Worry:** Persistent concern over stressors that may be real or imagined.

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- **Fear:** An intense emotional response to immediate threats, which can activate the fight-or-flight response.

Cognitive Reactions Under Stress

- **Concentration Difficulties:** Stress can impair the ability to focus, leading to decreased productivity and performance.
- **Confusion:** The overwhelming nature of stress can cause disorientation and indecisiveness.
- **Loss of Control:** A perceived inability to manage one's circumstances or outcomes.
- **Self-Esteem Issues:** Chronic stress can deteriorate an individual's confidence and self-worth.

Objections to Response Based Definition

A major problem with this approach is that these same responses occur as a result of very different stimuli (e.g., heart rate may increase as a result of physical exercise, viewing a horror movie, riding a roller coaster, or waiting for a blind date). Likewise, the psychological manifestations of anxiety, loss of felt control, and lowered self-esteem.

Stimulus-Based Definition of Stress:

Environmental events that impact on any of several response systems either immediately or as a result of prolonged exposure.

Other theorists have argued for a stimulus-based definition of stress. They argue for taxonomy of environmental events based on their covariation with systemic, psychological, and/or social disturbance, and include such events as noise, air, and water pollution, population density, odors, loss of loved ones, and changes in life style.

A stressful environment, for example, is one where noise level exceeds some specified decibel (dB) level, or where the carbon dioxide concentration equals or exceeds some agreed-upon value, or where environmental occupants exceed some number.

Problems with Stimulus Based Definition

It can be seen that a major problem with this view is that any stimulus may or may not be disturbing (stressful) depending on the personality of the individuals involved, the situation in which the event occurs, other competing behaviors, and the rewards and costs involved in dealing with the event. For example, a loud stereo can be enjoyable at a party, but very disturbing (stressful) if you are preparing for final exams; likewise, both high population density and isolation can be stressful or enjoyable depending on the amount of contact desired.

Lesson # 22

Stress As Cause and Effect

It is possible to think of stress as both something that is happening to a person and the person's response to what's happening. It involves environmental and psychological events, the interpretation of these events, and behavioral as well as physiological responding. Noisy environments, for example, may be related to physiological, psychological, and behavioral changes in those exposed to the noise. These responses, in turn, may change the nature and interpretation of the noise itself (i.e., noise changes neural activity in the reticular activating system, which subsequently changes the organism's perception of the noise; this altered perception, in turn, influences reticular activating system activity, and so on). **Stress, therefore, is neither the stimulus nor the response; it is a process involving both, and, as a process, it influences the ways in which environmental events are attended to, interpreted, responded to, and changed. It is also the process within which the responder also is likely to be changed.**

The Cyclical Nature of Stress

- **Environmental Stimuli:** Situations like noisy environments act as initial stressors.
- **Physiological Response:** These stressors trigger changes in the body's physiology, such as alterations in the neural activity of the reticular activating system.
- **Psychological Interpretation:** The individual's perception of the stressor is modified, which can either amplify or diminish the stress response.
- **Behavioural Adaptation:** The individual's behavior adjusts in response to the altered perception, potentially leading to a change in the environment itself.

While the specification of just what stress is has yet to be fully articulated, it is clear that such a thing actually exists, and it appears to involve physiological, psychological, and behavioral responding. It is also clear that at times it may even play a broader role by affecting human social systems. Confronted by environmental events which pose threat, challenge or danger, organisms respond physiologically, psychologically, and behaviorally. These responses not only are helpful in meeting the demands of the changing environment, but may even alter that environment, making it more benign (not always without cost to the organism).

Monat and Lazarus's Definition of Stress (1977)

They defined stress as any event in which environmental demands and/or internal demands (physiological or psychological) tax or exceed the adaptive resources of the individual, his or her tissue system, or the social system of which one is a part. Within this definition, change becomes stressful only when it strains the coping capacities of the entire organism or environment system to adapt to the change.

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This definition encompasses aspects of both the stimulus and the response, and includes the organism as an active participant in the process. It is therefore consistent with definitions utilized by most current researchers and is the one that will be used when referring to stress throughout the remainder of this text.

The Role of Perception in Stress and Adaptability

Perception plays a pivotal role in the relationship between organisms and their environment. Typically, the environment is perceived to be within the organism's "window of adaptability," allowing for a steady-state interaction where the organism can maintain equilibrium. However, the definition of stress acknowledges that this balance is not always maintained and that states of disequilibrium can arise, challenging the organism's adaptive capabilities.

Understanding Disequilibrium and Adaptive Resources

Disequilibrium occurs when environmental demands tax or exceed the adaptive resources of an organism or its environment. This can lead to stress, which is a state of tension resulting from adverse or demanding circumstances. The organism's perception of these demands plays a crucial role in the stress response.

Appraisal:

Appraisal is the process by which an organism evaluates the demands of the environment and assesses its own resources to cope with these demands. This cognitive evaluation is fundamental in determining the level of stress experienced and the subsequent response to it. Appraisal involves two key components:

1. **Primary Appraisal:** The process of recognizing and assessing the significance of an environmental stressor.
2. **Secondary Appraisal:** The assessment of one's resources and options for coping with the stressor.

Attitudes and Perception in Appraisal

Attitudes towards potential stressors play a crucial role in appraisal. For instance, individuals who view nearby airports as economically beneficial are likely to be less bothered by the associated noise compared to those who do not recognize such advantages. This illustrates how positive attitudes towards certain elements can lead to a downplaying of potential threats.

Types of Appraisals

There are three fundamental types of appraisals

1. Harm or loss assessments
2. Threat Appraisal
3. Challenge Appraisal

Harm or loss assessment:

It typically involves analysis of damage that has already been done. The properties of a sudden event such as a tornado may predispose people toward such appraisal because damage is done very quickly and people are more concerned with the

immediate consequences than the possibility of more. Harm-loss evaluation in the wake of the loss of a loved one, although when a loved one has been chronically ill for an extended period of time, bereavement may also occur in anticipation of loss. In addition to the actual loss of a loved one, we may be concerned with demands that will occur after death. Concern over these types of demands can be interpreted as challenges or as threats, that is, "Let's get on with life," or, "How will I get on with life?"

Threat Appraisal:

It is concerned with future dangers. If a tornado is sighted, it may initially be appraised as a threat and subsequently be appraised as something else. The stress of moving away to college, of learning to live with a roommate, and similar events is largely anticipatory as a student prepares to start school. Likewise, waiting to take an exam may be more stressful than taking it or even failing it. The ability to foresee problems and anticipate difficulties allows us to solve them or prevent their occurrence. At the same time, though, it may lead to the perception of threat and, thus, anticipatory stress.

Challenge Appraisal:

It focusses not on the harm or potential harm of the event, but on the possibility of overcoming the stressor. Some stressors may be beyond our ability to cope, but we all have a range of events with which we are confident of our ability to cope successfully. Stressors that are evaluated as challenges fall within this hypothetical range. The event may be seen as potentially harmful, but we feel that we can prevent the harm from occurring. A person may have just lost his or her job because of plant relocation. This stress can be seen as threatening (how are we going to make ends meet; how will we survive?) or as challenging (what else can I do to make a living; how can I make the best of a bad situation?).

Arousal

Arousal, refers to the physiological and psychological state of being alert and ready for action. It is a necessary component of the stress response, preparing the body to face potential threats or challenges.

Reticular Formation:

The reticular formation is a diffuse network of interconnected neurons located throughout the brainstem, extending from the spinal cord to the thalamus. It plays a vital role in controlling the overall level of consciousness and is involved in various functions such as regulating the sleep-wake cycle, filtering incoming stimuli to the brain, and maintaining alertness.

Reticular Activating System:

The Reticular Activating System (RAS) is a crucial neural network within the brainstem that extends to the midbrain and forebrain. It serves as a sentinel, filtering incoming stimuli and regulating wakefulness and consciousness.

The RAS performs two primary functions:

1. Information Screening:

It screens information on its way to the higher centers in the brain, blocking irrelevant information and allowing relevant information to pass upward where it can be processed and acted upon.

2. Alerting the Cerebral Cortex:

Activating the cerebral cortex into a state of alertness without specifying the nature of the alert.

According to Beck (1983) the reticular system works "something like a fire alarm that gets people into action but does not really say where the fire is". Through this system the organism is made vigilant and aware of what is happening in the environment, and is made ready for action. Included in this readiness are "increased metabolism of carbohydrates to produce more glucose and the release of fatty acids for greater energy, higher heart rate and oxygen consumption, constriction of blood flow to peripheral areas of the body with greater supply to the skeletal muscles, kidneys and brain" (Evans & Cohen, 1987; p. 576). This increased readiness, coupled with appropriate information about bodily needs and environmental demands, plays an important role in determining the ultimate expression of behavior.

LESSON 23

PHYSIOLOGY OF STRESS

We will begin lesson 23 by reviewing lecture 22 and further discussing the following topics related to Stress:

- Definition of Stress. (already discussed in detail in lecture 22)
- Cause and Effect. (already discussed in detail in lecture 22)
 - Appraisal (already discussed in detail in lecture 22) ➤ Arousal from the perspective of the Physiology of Stress.

Physiology of Stress and Arousal

Stress is a natural response to various stimuli in our environment, whether positive or negative. When faced with demands or changes in our surroundings, our bodies undergo physiological changes to adapt and respond effectively. These changes are orchestrated by complex systems within our brains and bodies. It is crucial to understand the physiology of Stress, as when we experience stress, our body undergoes various changes such as increased heartbeat and blood circulation. These bodily changes are reactions that prepare us to deal with environmental stressors. This process is commonly referred to as arousal. So, let's delve into arousal from the perspective of the physiology of stress.

Arousal

Arousal refers to a stirred-up state of mind in which we are inclined towards certain types of actions. In this state, we are either thinking about or being driven toward specific actions to address a scenario. Arousal can result from excitement, anger, or fear, signaling that our body is ready for action.

Adaptation: Hormones and Biological Systems

Arousal serves as our defense mechanism to cope with changes in our environment, preparing us to deal with significant and drastic environmental changes. Therefore, there must be a system within our body that triggers this entire mechanism. This system includes various organs and components, such as:

- Different types of hormones.
- Different types of organs in the central, peripheral, and autonomous nervous systems.

The nervous system components involved in this process include the hypothalamus, cerebral cortex, reticular formation, limbic system, and autonomic nervous system. These organs and systems activate to prepare us to address environmental challenges.

Reticular Formation

The reticular formation plays a crucial role in the physiology of arousal. Its primary function is to induce a state of readiness in the organism, indicating that the individual is prepared and quick to act. For example, before the start of a race, athletes assume their positions and await the starting signal. At this moment, they are in an increased state of readiness, with their bodies primed for action. Inside their bodies, the reticular activating system (RAS) functions are triggered, resulting in heightened metabolism

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rates and increased heartbeats to supply energy to the muscles. In such situations, blood flow is directed towards the muscles, while blood flow to internal organs like the kidneys or liver decreases. This physiological response prepares the athlete to address the scenario effectively.

Levels of Arousal

As arousal refers to the state of readiness or activation level of the body and mind in response to environmental stimuli. It can vary from low to high, with each level affecting our ability to perform tasks effectively. Understanding these levels is essential for optimizing performance and managing stress.

Increased Readiness: Extreme events or stimuli trigger a heightened state of arousal, characterized by increased heart rate, tremors, and heightened physiological responses.

Medium Level: This optimal state of arousal enables individuals to perform tasks efficiently and adapt to environmental demands without feeling overwhelmed or fatigued.

Lowest Level: A state of low arousal may lead to feelings of lethargy and disengagement, hindering performance and motivation.

The Role of Arousal in Performance

A crucial aspect of arousal is its impact on performance. While moderate levels of arousal can enhance performance by promoting alertness and focus, excessive arousal or stress can impair performance and lead to negative outcomes. Striking a balance between arousal and performance is essential for achieving optimal results.

Examples: Every time we experience stress, our body undergoes similar responses, entering a state of increased readiness. This heightened state prepares us to deal with actions in our environment. For instance, imagine driving along a main road when suddenly, a car speeds towards you, threatening a collision. Your reflexes kick in, and your body reacts swiftly to avoid the danger. In such moments, your heart races, and your body trembles. However, extreme readiness, akin to soldiers facing life-threatening situations, can also impair performance.

On the other side both extreme readiness and extreme lack of readiness negatively impact performance. Demoralized students lacking motivation represent one extreme, while soldiers facing imminent danger represent the other. Both scenarios hinder optimal performance. Consider a demoralized student who, with proper motivation, moves toward the middle of the readiness continuum. Here, motivation to study increases, leading to improved performance. Conversely, extreme readiness, akin to the state experienced during an imminent accident, hampers performance due to overwhelming emotion. Therefore, maintaining a balanced state of readiness is crucial for optimal performance, whether in academic or life-threatening situations. Maintaining a middle-level arousal is essential for peak performance, through maximum focus and concentration it can be achieved.

It is important to note that the Reticular Activating System (RAS) triggers these mechanisms, with readiness being both biochemical and psychological. On one side Motivating students to boost morale and move them toward the middle of the continuum is vital which is the psychological aspect. Additionally, mood significantly

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affects performance, highlighting the role of cognitive perception in behavior. On the other side, damage to the RAS leads to a comatose state, while drugs like amphetamines increase RAS activity and barbiturates depress it. So, it is important to study every aspect of arousal, whether it is neurochemical, psychological, or cognitive.

Hence, Arousal is closely related to stress and alerts organisms to changes in bodily needs and environmental demands. Understanding how arousal translates into goal-directed behaviors is also crucial for comprehending human behavior's influence by the external environment.

Arousal and Goal-Directed Behavior

We behave in certain ways to achieve our goals or objectives, and the environment plays a significant role in this process. For example, when entering a mall, the environment, such as billboards, induces an arousal state in individuals, directing them towards their goal, such as shopping.

General Adaptation Syndrome

Stress is not inherently negative; it is our adaptive reaction to the environment, enabling us to either change ourselves or the environment to establish a better equilibrium between the two.

Structures of Cerebral Cortex

The limbic system, housed inside the skull, comprises four important parts:

Thalamus: Relays information from sensory organs to the cerebral cortex.

Hypothalamus: Regulates feelings of fear, thirst, sexual drive, and aggression.

Amygdala: Influences motivation, emotional control, fear response, and interpretations of nonverbal emotional expressions.

Hippocampus: Plays a role in emotions, memory, and comparing sensory information to expectations.

In conclusion, our sensory organs process information, converting it into neurochemical energy that is transmitted to the relevant brain centers for processing. Additionally, glands, such as the pituitary gland, play a vital role in performing various bodily functions. Both systems are crucial for our body's functioning.

Pituitary Gland:

Situated within the brain, the pituitary gland serves as a crucial producer of human growth hormones and plays a pivotal role in regulating other hormone-producing glands. Referred to as the "master gland," the pituitary gland integrates incoming information from various sensory organs within the body. This information undergoes processing in the brain, which subsequently issues instructions. These instructions are then carried out with the assistance of glands, which produce different secretions to influence the functioning of internal organs.

The pituitary gland, as the master gland, not only activates other glands but also regulates the secretion of growth hormones, ensuring consistent growth from birth onwards. Growth is a fundamental physiological function that the pituitary gland

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oversees. Additionally, the pituitary gland controls the hormonal secretions of other glands, maintaining hormonal balance in the body.

The nervous system and glands work in tandem as a cohesive unit, preparing the body to respond to environmental changes and stimuli. This collaborative effort ensures that the body can adapt and function optimally in varying environmental conditions.

In the next lecture, we will study the General adaptation system in detail.

Lesson 24

General Adaptation Syndrome

In today's lecture, we'll delve into the intricate world of environmental psychology, focusing specifically on the physiological responses to stress and arousal considering general adaptation syndrome. Let's break down the key concepts that will be discussed in detail:

- Physiology of stress (Studied in lesson 23 in detail)
- General Adaptation Syndrome

Stimulus-Response Pattern: A Closer Look

When we encounter stimuli, whether positive or negative, our brain evaluates the information received through sensory organs. This evaluation triggers certain feelings and thoughts, leading to action. However, this process is not simplistic; it involves multiple stages, including stimulation, feelings, thoughts, and ultimately, action.

Hans Selye's Adaptation Syndrome

In 1976, psychologist Hans Selye introduced the concept of adaptation syndrome, proposing that stress unfolds in three distinct stages:

1. Alarm Reaction
2. Stage of Resistance
3. Stage of Exhaustion

1. Alarm Reaction

In response to any stressor, either physical or psychological, the hypothalamus is activated, mediating the secretion of large amounts of ACTH by the pituitary. This ACTH, in turn, stimulates the adrenal cortex to secrete increased amounts of adrenal corticoids. In general, these hormones activate the organism (as discussed in the previous section) allowing it to deal more adequately with its environment. This phase is called the alarm reaction.

2. Stage of Resistance

In the second stage, the organism recovers from the initial stress and begins to attempt to cope with the situation, mobilizing the body physically as well as psychologically to meet the demands of the environment. The organism, in a sense, is resisting the demands of the situation. This stage was, therefore, labeled by Selye as the stage of resistance.

3. Stage of Exhaustion

If the organism is unsuccessful in its attempts to cope, or if the stress persists, the stage of exhaustion is reached. At this stage the adrenal gland can no longer respond to the stress by secreting adrenal corticoids and the organism has exhausted its ability to cope with the stressor.

Important issues for consideration in this area are: how the stress of overpopulation is

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related to increased adrenal activity and to adrenal hypertrophy; how this has an inhibiting effect on gonadal functioning resulting in a decline in reproductive fitness, and ultimately to a decline in population and population density.

Evans and Cohen (1987) suggest that chronic exposure to a variety of environmental and/or social psychological conditions can in and of themselves be stressful, that the process of coping with them can be stressful, and that the "energy available for dealing with these conditions is limited. Thus, like Selye, they suggest that prolonged stress will eventually deplete the individual's adaptive resources.

Both approaches imply that stress produces a physical "wear and tear" on the system, and when coping abilities are exhausted, the individual becomes vulnerable to a variety of physical and psychological disorders. For further discussion and a comparison of the psychological and physiological models of stress see Baum, Singer, & Baum (1981), Cohen, Evans, Stokols, & Krantz (1986), and Evans & Cohen (1987).

Organisms, including humans, undergo stress in the context of the greater environment of which they are a part. At any given moment the environment/ organism system can be in a state of equilibrium on some dimensions, but in disequilibrium concerning others. It is therefore important to be able to identify potential sources of stress and to determine the levels of those sources likely to lead to disequilibrium.

There is growing awareness of the importance of psychological factors (i.e., cognitive, and emotional processes) in response to the environment. Additionally, such factors as beliefs, attitudes, and perceptual sets may, themselves, act as threats. For these reasons, researchers concerned with understanding stress are increasingly considering the impact of psychological variables and are incorporating them into stress theories as both mediators of physical stressors and as stressors (Frankenhaeuser, 1978).

Psychology of Stress: Primary and Secondary Appraisal

Moving on to the psychology of stress, appraisal plays a crucial role in shaping our response to stressful situations. **There are two types of appraisals:**

- 1. Primary appraisal**
- 2. Secondary appraisal**

1) Primary appraisal: Interpretation of threat

Primary appraisal involves interpreting the situation and assessing its potential threat. This initial evaluation influences how individuals perceive and react to stressors.

Appraisal of potential stressors depends on several factors including attitudes toward the source of noxious stimulation. Our attitudes and perceptions significantly influence how we perceive and respond to stress. Cognitive perspective also plays a crucial role, as our interpretation of a situation can either exacerbate or mitigate its perceived threat. For instance, individuals exposed to authoritarian parenting may adopt submissive attitudes towards authority figures, while those encouraged to express their talents and opinions may adopt assertive approaches.

2) Secondary Appraisal: Selecting Coping behaviors and evaluating their effectiveness.

Secondary appraisal occurs when individuals assess their resources and coping strategies to manage the stressor. They weigh the potential costs and benefits of different responses and choose coping strategies based on their past experiences.

Lastly, our ability to cope with stress relies on various resources, including:

- Psychological
- Social
- Genetic factors

These resources encompass our personality traits, social support networks, and genetic predispositions, all of which influence our resilience in the face of stress.

In the next lecture, we will discuss other areas of stress.

Lesson 25

MEASURING STRESSORS

In the last lecture, we discussed General Adaptation Syndrome and the Psychology of stress while in this lecture we will talk about:

- Hens' adaptation model (discussed in detail in lecture 24)
- Researching and Measuring Stress

Researching Stress

Studying stress scientifically is crucial for gaining insights into its nature and effects. One of the primary objectives in this endeavor is to understand how to effectively measure stress. To achieve this, it is essential to first establish a clear definition of stress. By defining stress, we can identify various aspects or components that can be measured, thus facilitating accurate assessment of stress levels.

Stress in environmental context: Organisms undergo stress in the greater context of their environment. Environment impacts the individual, either its people around them or other environmental elements that affect them. Organisms stay in balanced condition with the environment. But this balance does not always stay as change is the only constant. This imbalance state is considered as stress/distress. In case of this disequilibrium of how we react, our primary and secondary appraisal helps us to measure it.

Measuring Stress

There are two perspectives on measuring stress:

1. Qualitative differences
2. Quantitative differences

Qualitative differences

Qualitative measurement of stress involves assessing the subjective experience and perception of stressors. Unlike quantitative measurements that focus on numerical data, qualitative measurements delve into the qualitative aspects of stress, including emotions, feelings, thoughts, and coping strategies.

To measure stress qualitatively it is important to classify the sources of stress. There are the following sources of stress:

1) Stressors

Stressors can vary in duration and intensity. Daily hassles, cataclysmic events, major life changes, and ambient stressors all contribute to the overall stress experience for everyone.

2) Conditions

One thing for a person can be pleasant in one condition and unpleasant/painful in other

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conditions/circumstances.

For example: Loud music is a great source of stress during studying and pleasant in the time of parties.

3) Chronic Stressors vs Acute/Recurring Stressors

Chronic stress persists over a prolonged period with lower intensity, while acute stress is characterized by higher intensity but shorter duration.

For example: Pollution is the source of chronic stress around us as it increases gradually with low intensity but for a prolonged time while in the parallel universe, this pollution is the source of acute stress for a person who belongs to hilly areas where there is no pollution and when they visited big cities it is the source of acute stress for them they perceive it with high intensity and for short period.

4) Controllable vs Uncontrollable

Few stressors can be controlled and there are few which cannot be controlled. For example, the temperature of the air conditioner can be controlled in your house but while traveling in a bus it cannot be controlled which causes flu or sore throat for some people.

So, stress can also be measured in the aspect of control as we can adjust the stressor according to our needs.

Evans and Cohen's typology of stress

Evans and Cohen compiled a typology of stress in 1987. There are 3 dimensions of this typology:

- Duration
- The magnitude of the required response
- Number of people affected.

In light of these dimensions, there is a need to study a few daily life stressors that are the sources of stress.

Daily Hassles

The category that includes some of the most chronic environmental stressors has been labeled *dailyhassles*. These are present during most of our daily lives and include such conditions as job dissatisfaction, neighborhood problems, crowding, pollution, and noise (Glass & Singer, 1972). For example, the daily life of a student from going home to school and coming back comprises a lot of stressors every day but when we talk about whether these stressors are chronic or acute then we can say that these daily life hassles are chronic as they build up gradually but for persistent time and we get used to it.

Cataclysmic Events

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Great natural upheavals that turn down our lives are called cataclysmic events, for example, Earthquakes or Floods. Their impact is huge and affects every person.

Major Personal Life Events

A third group of stressors includes those events powerful enough to individually challenge our adaptive abilities. These events include illness, death, or significant loss (psychological or economic).

Ambient Stressors

Environmental background conditions represent relatively stable, continuous, and intractable conditions of the environment and are labeled as ambient stressors such as work overload, job, poverty, family conflicts, and air pollution. They represent the relatively continuous, stable, and intractable conditions of the environment (Cambell, 1983).

Noticing Stress

The main question in the stress part is now when people start noticing stress. Stressors may go unnoticed until they interfere with some important goal or directly threaten one's health. For example, environmental pollution.

Stress and Predictability

The predictability of stressors also plays a significant role in how individuals respond. Regular stressors allow for better preparation, while unpredictable stressors can intensify the stress response.

Quantitative differences

Measures stress quantitatively physiological responses, help as our body helps to deal with the situation, and in that case physiology of our body changes. To measure stress quantitatively 4 elements are most important.

- Intensity
- Duration
- Rate
- Controllability

Intensity: The intensity of the stressor refers in some sense to the "power" of the stimulus. Intensity can be "measured," for example, in terms of the magnitude or frequency (loudness or pitch) of sound, the concentration (parts per million) of a population, the velocity of the wind, or the temperature or relative humidity of the atmosphere. It can also be measured in terms of its physical effects; for example, the number of homes destroyed in a flood, the number of persons killed by a tornado, the number of victims hospitalized because of a nuclear accident, or the number of birth defects resulting from water contamination.

It is generally assumed that the greater the intensity of the stressor, the greater the resulting

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stress response.

Rate. While some stressors occur once, others may be recurring. Their periodicity can be regular or irregular, predictable or unpredictable, and short-phased or long-phased. For example, the temperature of every month in these we can find a pattern that summer comes once a year.

Duration. Independent of the intensity of the stressor is its impact concerning time. The immediate presence of a tornado, for example, is relatively short (perhaps only a few minutes), but its intensity is great. On the other hand, an airborne pollutant (such as asbestos) might have a low concentration yet be ever-present. Duration and intensity, it can be seen, are independent dimensions. All other things being equal, we can expect stressors of long duration to have a greater effect on the stress response than stressors of short duration. However, the accumulative effect of low-level stressors over long periods may result in deceptively severe consequences. Any consideration of the potential effects of an identified stressor, therefore, must consider both the duration of the stressor and its intensity.

Controllability. Stressors vary concerning the degree of control humans have over them. It is possible to exert some control over the temperature of our indoor environments, over the noise level of our offices, and even over the traffic we must drive in. We do this by turning the thermostat either up or down, by putting sound-damping or sound-absorbing equipment in our offices, and by choosing the times of the day and the routes to take to avoid traffic. It is not possible to exert control over other stressors. We cannot stop hurricanes; we cannot prevent earthquakes. There are still other stressors where it is possible, theoretically, to exert control, but practically we are unable to (e.g., the loud stereo in the adjacent apartment or the litter in the streets below). Researchers have shown that the lack of control over environmental stressors where it is possible to have it can exacerbate the stress response.

In our discussion of environmental stressors throughout this text, we will, again and again, be looking at potential stressors in terms of their intensity, duration, rate, predictability, and controllability, and we will be calling for additional means of measuring these characteristics.

In the next lecture, we will discuss the immediate response and long-term Response.

Lesson 26

Measuring Stress

In this lecture, you will study about following

topics: •Psychological and Somatic Responses

- Measuring Organisms Immediate and long-term Responses
- Models of Adverse Health

Psychological and Somatic Responses

Psychological responses can pertain to our feelings, attitudes, cognitions, and behavioral patterns because psychology is the study of mind and behavior. Moreover, psychological responses affect somatic patterns. When we pass through stress our body must go through many changes to prepare itself and many physiological reactions take place. And if the stress is short-term, then the physiological reaction will be for a short period. Moreover, if these responses are for the long term, then they can hamper activity and create cause kind of danger to the body.

Measuring Immediate Response to Stress

It means to measure galvanic skin responses, the level of hormonal secretions, and measure level of secretion in blood regarding adrenaline and non-adrenaline (which are two hormones that help us to deal with emotional states).

Long term Response

We have little control over long-term stress and must live with it for a longer time. For example, people living in refugee camps.

When the condition of stress persists then these physiological reactions also continue for longer periods and this leads to certain kinds of physiological diseases.

Bio-medical Model

To understand disease bio-medical model tells us that there are 3 ways for a person to get ill.

- Illness results from a specific physiological dysfunction.
- Invasion of foreign substances.
- Some diseases do not fit the traditional bio-medical model and are named diseases of lifestyle (heart diseases seem to be correlated to the patterns of lifestyle).

Adverse Health Models

Dohrenwend & Dohrenwend Victimization: Model A

Dohrenwend & Dohrenwend (1981) have conceptualized the possible processes

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where stress induces adverse health changes (see Figure 5-4). In this figure, Model A contains no intervening processes. Rather, it postulates a simple and direct effect of stressful events on adverse health changes. It further suggests that the effects of stressful life events are cumulative, thus accounting for studies of extreme situations, such as combat or incarceration in concentration camps. Other severe stresses over which an individual has no control, such as the death of a loved one, are also accounted for by this model. This model is called the *victimizationhypothesis*.

Model B: Stress and Strain Hypothesis

Model B, the *stress-strainhypothesis*, postulates that psychophysiological strain mediates the impact of life events on subsequent health and illness. Evidence in support of this hypothesis comes from studies that show that if the effects of symptoms of psychophysiological straining are eliminated, correlations of stressful life events scores and measures of illness are significantly reduced (Bloom, 1985). This model is most closely related to the conceptualization of stress in this text.

Model C: The vulnerability!

The *vulnerabilityhypothesis*, suggests that there are preexisting personal dispositions and social conditions which moderate the causal relation between stressful life events and health. It is this model that suggests to researchers a search for such mediating factors as the strength of social support systems, and personal conditions.

Model D: additiveburdenhypothesis!

By contrast, postulates that personal dispositions and social conditions make independent causal contributions to the occurrence of pathology. This model suggests that personality variables and/or social conditions are a potential source of added burden to the individual in the precipitation of illness and is thus called the *addedburdenhypothesis*.

Model E: Chronic burden!

Proposes that transitory life events have no role in precipitating illness, but rather that stable personal characteristics and social conditions by themselves cause adverse health changes. This model is called the *chronicburdenhypothesis*.

Model F: Event Proneness Hypothesis

Finally, Model F portrays stressful life events as exacerbating already existing health disorders. It is called the *eventpronenesshypothesis* because stressful life events are thought to characterize individuals who are already ill.

Measuring Stress Psychologically

When we talk about psychological assessment, we note down all factors that contribute to behavioral patterns. Psychological measurement of the stress response has focused on various psychiatric symptoms and has utilized a variety of standard psychiatric symptom inventories. The common denominator among the various scales seems to be the extent to which respondents perceive themselves to be in an intractable situation marked by negative affect and a degree of uncertainty.

Measures of Coping

The level of stress a person experiences and the extent to which deleterious effects occur because of exposure to a stressor will depend, in part, on how well the individual copes with the stressor. Whereas Behavioral coping responses can be categorized according to individual differences. Every individual will cope with stressful events differently.

In the next lecture, you will study moderators of stress response.

Lesson 27

MODERATORS OF THE STRESS RESPONSE

After a review of the previous topics, lecture 27 will comprised of the following: ► Moderators of Stress Response

Moderators of Stress Response

By response, it means that a person is pitched against a stressful scenario and what type of response comes out of him. This response pattern is important to understand and study for psychologists to find out how they can build up a person's capacity even further to deal with the stressors. **The extent to which a stressor produces a response is moderated by many factors:**

- Attitude toward the source of stress
- Control
- The Hardy Personality
- Social support
- The Relaxation Response

All these factors will increase or decrease a stressor's impact, we turn now to a discussion of them.

1) Attitude toward the source of stress

As mentioned earlier, responses to stressors are necessarily related to how a stressor is perceived. Attitudes toward the source of the stress are important psychological factors and act as filtering devices that moderate the perception of the stressor.

Example: Let's consider the case of a student who may not perform well in his mid-term exams. This suggests that they are struggling to align their study routine with their academic goals. Despite making efforts, they find it challenging to adapt. At this juncture, it's crucial to engage with the student and understand their perspective on their academic routine. Sitting down with them, we can inquire about the difficulties they face in their studies. Perhaps they find attending morning classes challenging, or they feel overwhelmed by peer pressure, or they perceive the course material as too difficult. By assessing various stressors that students encounter at school, we simultaneously delve into their perceptions of these stressors. Do they believe they can overcome these obstacles, or do they feel incapable?

The student's perception holds significant weight. If they express a belief that they can't meet academic requirements, it's essential to connect with them on an emotional level and understand their underlying concerns. Recognizing their mental state, we can work on bolstering their confidence and resilience.

Once the student develops a positive outlook towards their academic challenges, we can then identify specific problems and work towards solutions. Now Let's also discuss phobias in the light of perception. **Phobias** are instances where individuals feel intense

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and irrational fear toward specific objects, situations, or activities. These fears can vary widely among people, but certain phobias, such as the fear of traveling, are more commonly observed.

It's important to note that phobias stem from irrational fears, and individuals' perceptions of these fears play a crucial role. For example, someone with a fear of flying may harbor negative thoughts and beliefs about air travel, leading to avoidance behaviors. When faced with their phobia, individuals may experience physical symptoms like rapid heartbeat, shortness of breath, and nervousness. These reactions stem from the negative perception they hold towards the source of their fear.

Addressing phobias involves shifting these negative perceptions towards a more positive outlook. Psychologists often employ techniques to gradually expose individuals to their fears in a controlled environment, helping them build resilience and overcome their phobias. For instance, individuals may be guided through relaxation exercises while imagining themselves in a feared situation, gradually desensitizing themselves to it. This process aims to move individuals from extreme fear (10 on a scale) to a more manageable level (1).

Therapists may also accompany individuals on real-life exposures to help them confront their fears directly. By addressing the negative perceptions associated with phobias, psychologists work towards empowering individuals to regain control over their lives.

So, to these above examples, people's perception of stress is very important, and it is also vital to control it.

2)Control

Another crucial aspect to consider is the perceived control over the stressor. The level of control one feels they have over a stressful situation plays a significant role in how they respond to it. When individuals feel they have some control over a stressor, they are better equipped to cope with it. However, when they perceive the stressor as beyond their control, it can become overwhelming. A poignant **example** can be found in the experiences of prisoners of war during World War II. Held in Nazi concentration camps, these individuals faced unimaginable hardships and lacked control over their circumstances. This lack of control contributed to chronic stress and various psychological disorders among them.

Similarly, in discussing phobias, the perception of control is crucial. If someone believes they can control their fear, it's possible to gradually build their resilience to it. Conversely, if they feel powerless/uncontrollable against their fear, intervention is necessary to address this perception.

Moreover, an **individual's overall physical fitness** level plays a significant role in their ability to withstand stressors. Physically fit individuals possess higher energy reserves, enabling them to endure challenging situations for longer periods. Conversely, those lacking physical fitness may find themselves more vulnerable to stressors, as their energy and resilience diminish.

Note: Understanding stress from multiple angles is essential. However, it's important to note that discussing stress does not aim to make individuals more vulnerable to it. Instead, it underscores the importance of resilience in navigating stressful circumstances. Ultimately,

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how one responds to stressors determines their success or failure in overcoming them.

The Hardy Personality

In the realm of leadership, the concept of a hardy personality emerges as a significant factor. Think about your favorite leaders. They could be historical figures like Muhammad Ali Jinnah, renowned poets like Allama Iqbal, or even sports icons who led their teams to victory. What sets these individuals apart as heroes?

When we envision a hero, we often picture someone who confronts daunting challenges with unwavering determination. Regardless of the adversity they face, they remain steadfast in their commitment to their goals. This resilience stems from their strong character, refusing to succumb to the pressures of their circumstances. Many heroes throughout history faced setbacks and failures, yet they persisted in their pursuit of their objectives. Some even sacrificed their lives before achieving their goals. Their heroism lies not in their immediate success, but in their unwavering resolve to overcome obstacles.

Consider the example of those who fought for independence or social justice. Despite enduring hardships and setbacks, they remained dedicated to their cause, inspiring future generations to continue their struggle. In this context, a hardy personality emerges as a beacon of hope amidst the challenges of stress and adversity. Hardy individuals possess a sense of resilience and determination that enables them to navigate difficult circumstances without losing sight of their objectives. The key components of a hardy personality include a strong sense of commitment, a perception of challenges as opportunities for growth, and a belief in one's ability to exert control over their life. When faced with setbacks or failures, it's natural to experience feelings of selfdoubt or frustration. However, reframing these experiences as opportunities for learning and growth can help mitigate negative feelings.

Next time one encounters a setback, consider redefining it as a learning opportunity rather than a mistake. By adopting this mindset, we can cultivate resilience and develop the resilience needed to overcome challenges. Consider reframing mistakes or challenges as opportunities for growth. Every problem you encounter presents a chance for personal development. While it's natural to feel daunted by challenges, approaching them as opportunities allows you to harness your intellectual and mental capacities. View each problem as a canvas upon which you can paint your journey of growth. By tackling challenges head-on, you not only overcome obstacles but also enhance your skills and resilience. Reflect on your experiences and mistakes as opportunities for learning. Embrace the equation: Mistake plus learning equals growth. Every error becomes a stepping-stone towards personal development when accompanied by insights gained from experience.

Conversely, failure to learn from mistakes stuns growth. Remember, mistake minus learning equals stagnation, a state antithetical to personal progress. Understanding the concept of hardiness is essential. It encapsulates resilience and determination in the face of adversity, forming the cornerstone of personal growth. As we conclude this discussion, remember to approach life with an open mindset, embracing challenges as opportunities for growth.

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In our next lecture, we'll delve into our next topic.

Lesson 28

Environmental and Cultural Variances

In this lecture, we're delving into how the environment and culture influence human behavior, and how human behavior, in turn, shapes the environment and culture. Here's an overview of what we'll cover in this chapter:

- First, we'll examine the intricate relationship between behavior and environment, summarizing how they interact with each other.
- Next, we'll explore phenomena like conformity and obedience, including a case study to illustrate these concepts.
- We'll then analyze the factors influencing conformity and individual behavior within groups.
- We aim to understand how individuals conform to certain ideas, become obedient, and adapt to different environments.
- Moving on, we'll discuss attributing behavior to both personal dispositions and situational factors. This section will help us grasp how people interpret and attribute behavior.
- A detailed exploration of group behavior follows, covering topics such as group pressure, conformity, and the reasons behind group influence and mob behavior.
- Our final focus will be on culture. We'll investigate how the environment shapes culture and how individuals learn about and adapt to the cultures they inhabit. Key processes like enculturation and acculturation will be explored, along with the roles of language, symbols, and rituals in cultural formation.
- We'll view culture as a dynamic force, constantly evolving and influencing individuals and society. Finally, we'll touch on the measurement of culture and briefly discuss subcultures.

Relationship between environment and human behavior

We've come to grasp the intricate relationship between human behavior and our surroundings. Our environment can significantly shape our actions, sometimes to the extent that we might attribute our behavior solely to external influences. Yet, we mustn't overlook the undeniable impact of the environment on our choices and actions. This dilemma echoes the age-old debate: is it the environment or human free will that predominantly governs our behavior, or is it a complex interplay of both? There are instances where individuals act in ways they never anticipated, highlighting the profound influence of environmental factors. It's a nuanced interaction between the environment and individual agency that ultimately molds human behavior, posing a perpetual conundrum for psychologists and philosophers alike.

Human behavior often defies expectations, showcasing a remarkable adaptability to various situations and environments. Consider individuals who typically exhibit calm and peaceful demeanors suddenly becoming aggressive under certain circumstances, or those who are naturally aggressive displaying unexpected submissive behavior. Moreover, there are instances where introverted individuals become talkative and extroverted in specific situations. These examples underscore the significant role

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played by environments and situational factors in shaping behavior. Yet, amidst this complexity, the question persists: is it ultimately the exercise of human free will that guides behavior, or does external influences/environment wield greater control?

To understand this concept more deeply a case study will help us.

My Lai Massacre Case Study

In March 1968, three platoons of American soldiers, non-collectively as Charlie Company, swept into the South Vietnamese village of MyLai, and killed several hundred unarmed civilians, women and children, and old men. Most of the soldiers joined in shooting those who did not fire, not one attempted to stop the slaughter. Some of the men of the Charlie Company reported that they followed the orders of their leader, a lieutenant of the US Army.

Others said they simply followed the example of others. The gruesome, isn't it? The question is, why would the soldiers behave so savagely? There are certain other details. The soldiers of the Charlie Company had joined in Vietnam about two to three months before this incident took place. They mentioned that they were being constantly attacked. They were fighting a guerrilla war over there because they were in the enemy area, and there was insurgency taking place over there. They were under extreme stress, of course. And they received the orders that in the My Lai village, the American army suspected the armed men of the Vietnam army, and they had to launch an operation to go there and get those men. And they simply went in there. Just like one of the soldiers said, "I was just fulfilling my orders over there".

However, the types of pictures that we have seen are extremely gruesome, because there were few men, and all those men were old men, and there were ladies and children. If the troops had to enter that village, the question is, why would they behave so savagely?

We understand that the training of the army is very, very extensive. This is not only physical training, but this is also a lot of mental training. Soldiers are raised to become civilized citizens. Thou, their job is to fight wars, is to defend against the enemies of the country. However, their training is made sure that they learn a lot about how to become civilized citizens. The question is that any person who enters such sort of a situation would be behaving the way the American soldiers behave. Because normally, ordinarily, normal people do not tend to see these types of situations. However, when we come to think of it, can you say that we can forgive those soldiers because they were under extreme stress? See, it is a question that relates to the same debate. Was it the environment? Was it the situation? Or was it the free will of the people? Students, we have some more information about this my life massacre. However, before that, we would like to tell you what one of the Charlie Company soldiers said. He said, I looked around and saw everyone shooting. I did not know what to do, so I started shooting. This sounds more like this soldier was under the grip of the circumstances. The environment was impacting upon him very, very heavily. And what he did, he did what the rest of the people were doing. Environment and behavior. Social environment and behavior. This means the social environment and the social dynamics of that place at that time just prompted the soldier to do the same way as the other members of the group were doing. So, is it a justification?

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Let us give you some questions to think about this massacre. The question is, is the way people think and act the production of their inner dispositions? If you take this question to understand the My Lai massacre, the question would be, do you think what people did over there? What Americans would have stood over there was based upon their inner disposition because they were the people like that. Or because they were in that situation where somehow the circumstances made them behave in a certain way. The next question is, does the situation play a role? Was it only because of the situation that soldiers behaved savagely the way they did? The next question is, do environmental factors contribute to the way people think, feel, and act? The way those soldiers were thinking, feeling, and acting, was because of the environment. Was it because of the situation that made them do that? And finally, the question is direct, why did the soldiers in my life behave savagely? Why did the people behave very, very savagely over there?

Of course, as psychologists, we can look at this phenomenon and we can bring in all the technology, the behavior technology that we have learned to understand this entire fiasco of dreadful human behavior. However, let us give you some more information on the My Lai massacre. The remaining information is something like this. While the massacre was taking place, a helicopter happened to be passing by that village. This is what this all. A helicopter pilot from an aerospace team witnessed many dead and dying civilians as he began flying over the village. He landed in his helicopter and asked the surgeon he encountered there if he could help get the people out of the ditch and the surgeon replied that he would help them out of their misery. Later, he told his crew that if the US soldiers shot at the Vietnamese while he was trying to get them out of the bunker, they were to open fire at these soldiers. Now, students, this is entirely antithetical behavior to what was going on from the soldiers of the same army. From the same army, few soldiers were engaged in the massacre while from the same army, there came a few other soldiers who did not want those soldiers to engage in that massacre and they were so charged for that that they would even fight against their fellow soldiers. These soldiers were later given medals for their bravery.

Once again, the question to you is? these two divergent types of behavior Is the environment playing upon these people to behave in two different ways or is it the people themselves that are trying to interact with the environment or they are trying to control the environment in different ways? Once again, let me remind you what that soldier said. I looked around and saw everyone shooting, I didn't know what to do so I started shooting. Is that enough reason for that? Of course, this is a question that you would be thinking. At this point students and based upon this soldier's interview and what he said in that, I would like to give you the definition of conformity.

Conformity

Conformity means a tendency to shift one's opinions or actions to correspond with those of other people because of implicit or explicit social pressures. Conformity is a phenomenon here. Conformity is pervasive in society, influencing career choices, among other things. For instance, in our country, many gravitate towards careers like medicine or engineering, following societal norms.

Why do people conform? Often, it's to align with prevailing standards. We may sacrifice our true desires to fit in. How many aspire to less conventional paths, like philosophy or poetry? Society molds our aspirations, nudging us towards conformity.

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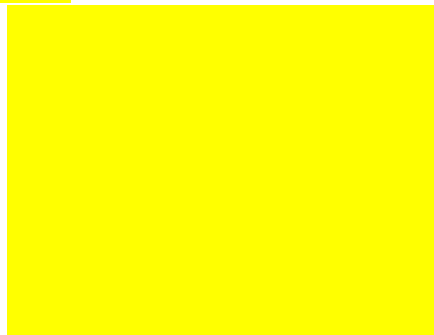
Ultimately, conformity reflects the tug-of-war between societal norms and individual autonomy. It's a complex dynamic that shapes behavior in profound ways.

Let's revisit the My Lai case and the question of why the soldiers behaved savagely. Similarly, let's consider why young people often conform to societal norms when choosing their careers. It's tempting to label the soldiers as bad or the conforming students as lacking in courage.

Asch's Experiment

But let's examine how people conform to social norms through a simple experiment.

In this experiment, subjects were shown two sets of lines and asked to judge their lengths while working in a group. Seven subjects participated, but only one was real—the rest were confederates instructed to give obviously wrong answers. Despite the subjects being able to clearly see the correct answer, 37% still conformed to the group's incorrect responses.



Factors Influencing conformity.

There are the following factors that influence conformity:

Agreement amongst participants

Dissenters

Anonymity

1) Agreement among participants

Why did they conform? This experiment, known as the Asch experiment, sheds light on the factors influencing conformity. One key factor is the agreement among participants. When everyone else in the group gives the same wrong answer, individuals often succumb to groupthink, even if it contradicts their perceptions.

For 37% of the participants, the pressure to conform outweighed their ability to trust their judgment. This highlights the powerful influence of social norms on individual behavior.

Let's revisit the scenario of the American soldier in Vietnam who said, "I looked around, everybody was shooting, so I started shooting." Similarly, imagine a conversation with a young person today who chooses a career path based on societal influence, like, "My dad said computer scientists make great money, so I'm going for it. Plus, lots of my

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friends are studying computer science." In both cases, individuals are following the crowd, conforming to social norms. **2) Dissenters**

Now, let's examine this from a different angle. Think about dissenters—those who go against the group's thinking. Picture a scenario where a group is unanimously leaning towards one option, like going to a certain place, but one person suggests an alternative. This dissenter challenges the group's consensus, opening new possibilities. Research, including the famous Asch experiment, shows that dissenters can significantly reduce conformity rates compared to groups without dissenters.

3) Anonymity

Another factor to consider is anonymity. When individuals feel anonymous, they may express their true opinions more freely. In experiments where subjects remained anonymous in aloof rooms, they were less likely to conform, indicating the impact of social pressure on conformity.

4) Self-confidence vs self-doubt

Additionally, self-confidence versus self-doubt also plays a role. Confident individuals are more likely to resist conformity, while those plagued by self-doubt may succumb to group pressure more easily.

Reasons for conformity

What is the reason that moves people to conform with the group? One of the great reasons is often involves people outwardly agreeing with the group while inwardly retaining personal opinions. This internal conflict can be stressful, leading individuals to eventually adopt the group's viewpoint to reduce discomfort—a process known as internalization.

Process of Conformity

- In confirming people first go through the process of Identification. **Identification** means they want to go along with the group.
- In the second step, they go through the process of Internalization. **Internalization** comes when they start believing in the group's point of view. They kill their own point of view.

Now, Consider historical examples like World War II and the rise of the Nazi party. Despite initial resistance, many individuals eventually conformed to Hitler's ideology due to social pressure and the suppression of dissenting voices due to group norms. It was also because dissenters were not treated rightly.

Ultimately, the interplay between environmental factors and individual agency shapes behavior. While social influence can be powerful, individuals with strong convictions and confidence may resist conformity, paving the way for dissent and change. So, is the environment more powerful, or is human will? It's a question worth exploring scientifically, as we seek to understand the complexities of human behavior.

In the next lecture, we will study Attribution Theory.

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Lesson 29

Attribution Theory

In this lecture, after reviewing the previous lecture you will study the following topics:

- Attribution Theory
- Attribution Errors
- Internally vs Externally caused Behaviors.

Attribution Theory

Our perceptions and judgments of people's behavior are heavily influenced by our assumptions about their internal state, a concept known as attribution theory. For instance, we might come to start thinking that this gentleman is a person who is a bad guy because he is trying to demean us, or he is trying to embarrass us in front of the crowd. Or a certain person comes to us, and he just becomes friendly, gives you a nice shake, introduces himself, and starts talking to you. You might say that he is a nice guy because he is behaving like a nice guy. And when we look at both people, we try to see what is the state that is inducing this type of behavioral pattern. If you come to find out that this type of thing is being done by the person because he internally actually plans to do that to you, then you would consider that person bad. However, if you come to think that that person was doing, just throwing a joke to make everybody laugh at the party and make you laugh also at the party, then you might come to interpret the same thing from a very different point of view.

Attribution theory is a theory that helps us to understand 2 things:

1. How do we explain people's behavior
2. How do we come to attribute people's behavior to their internal states

Let's consider an **example**. Syra, a student of business administration, seems shy in class but talkative with friends. Is she inherently shy, or is her behavior situation-dependent? If we observe her only in class, we might conclude she's shy. But if we see her with friends, we might see her as confident. This illustrates how our judgments can be influenced by the context in which we observe behavior.

Attribution theory has been developed to explain why we judge people differently based on the meaning we attribute to their behavior. We attempt to determine whether behavior is internally or externally caused. It's crucial to consider whether someone's actions stem from their personality or the situation they're in.

When someone's behavior is internally caused, we hold them responsible for their actions. However, in some cases, external factors can influence behavior. For example, someone might behave differently in stressful situations. It's essential to consider these factors when making judgments about people's behavior.

However, as we discussed in the last lecture, sometimes it happens that people in certain situations behave in a queer sort of way. In one situation they are cool and calm. And when you capture them in an aggressive mode, you might come to say that this gentleman is very, very aggressive. However, you need to see that aggressive behavior is internally caused or externally caused. If that behavior is externally caused according

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to your perception, then you might say that this gentleman is behaving aggressively only in this situation. However, if you come to think that this behavior is internally caused, then you would say that this gentleman is aggressive. In either case, our perception might be wrong because you do not have complete information on that.

Attribution Errors

There are 2 types of attribution

errors. ➤ Fundamental

Attribution error

- Self-Serving Bias

□ Fundamental Attribution Error

Fundamental attribution error means that we are always making wrong judgments based on our assumptions. Understanding the influence of external factors and overestimating the influence of internal factors. That is the fundamental attribution error.

For example, assuming that a person is late all the time because he or she is not interested in the work rather than finding out that the shift starts when the parking is full. We assume that something is wrong with this person as he is always late which means that the behavior is internally caused. However, there could be a major reason for that person to be late every day.

However, instead of finding out why this person comes late every day and helping this person out, we might simply build up an assumption without getting enough information that this person is a habitual latecomer. This person is lousy, he just wants to drift away from the responsibility. So, fundamental attribution error is an error that we commit when we try to underestimate the influence of external factors and overestimate the influence of internal factors. So, we underestimate and overestimate the influence of internal and external factors. This is the fundamental attribution error that we are constantly making.

□ Self-serving bias

Self-serving bias means that we attribute success to internal factors and failure to external factors. Like they say, nothing succeeds like success and failure has many excuses. When a person is successful, that person is successful and that means that success becomes part of that person and we do not say that this person is successful because of certain external factors. We simply say that that person is successful, and we come to think that that person might be very, very intelligent. So self-serving bias is attributing success to internal factors and failure to external factors. We might come to say that I failed because I faced these difficulties and if you are successful, we would say I am successful because I worked very, very hard for it. So, we try to tailor the opinion according to our choice. We try to tailor our perception of the situation in such a way that we appear to be heroes in that situation and this attribution fundamental attribution error is called self-serving bias.

Internally VS Externally Cause Behaviors

We believe that internally caused behaviors are under an individual's control. Externally caused behaviors are motivated by external forces. So, when we say that someone has done something because of his intent, then we tend to attribute the entire blame to that person. We would say that this person behaved in this way because this person is bad, or this person is good, or whatever that person is. However, when we find out that there are reasons for a certain behavior that is externally caused, we attribute the behavior to external factors, and this is the time that we usually try to forgive people.

So, we are trying to determine and one of the most important things out of these is that

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when we are trying to find out that the behavior is internally caused or externally caused is distinctiveness. As, how we determine the source of behavior is determined by three factors.

- Distinctiveness.
- Number two is consensus.
- And number three is consistency.

Distinctiveness

Distinctiveness means How much a person's behavior is unique. How much a person's behavior is distinctive? "Distinctiveness refers to whether an individual displays different behavior in different situations". If the distinctiveness is high, if the behavior is very, very distinct, then your attribution towards that behavior would be very high towards external factors. If the distinctiveness is low, then your attribution towards the external factor would also be very, very low because this behavior is not distinct at all. As we discussed in a previous **example**, if the employee comes late one day, then that behavior is very, very distinctive because it does not happen all the time. If the employee is coming late every day, then that behavior is not distinctive at all because this is something that happens every other day. And if that happens every other day, then that means you might come to say or attribute his behavior to internal factors. So, if we see that the person's behavior is distinctive, then we may say that this is externally caused. If this person's behavior is not distinctive, we may say it is internally caused. And internally caused means the person is penalized. **Consensus**

"If everyone who is faced with a similar situation responds in the same way, we can say the behavior shows consensus".

Example: If most of the students in a class let us say 90% of the students in a class fail in mathematics. Who is responsible for that? Is it the students or the teachers? Just make a general understanding of it. Just pass on the judgment right now 90% of the students in a class fail in mathematics. Is there something wrong with the students, or is there something wrong with the teacher? Based upon the consensus point of view, we will say that since 90% of the students are failing in the class, then that means something is wrong with the teacher because everyone is failing in this class. Hardly anyone passes. So, what are the teachers teaching in this class? Maybe there is something wrong with their teaching methodology. We see a consensus in most people's behavioral patterns, and then we say that this is understandable. You might have heard a lot of people who come say that I could not progress in life because I was born into a very poor family and my parents could not send me to a good school. And based on the poverty, I could not progress in life. And then you would say that most of the people who are in poverty are not able to progress in life. So, is it poverty that is responsible for their failure, or is it the people themselves who are responsible for their failures? Who is at the helm of the affairs? Poverty or the people themselves? Is it poverty that makes people fail, or is it the failed people who produce poverty? This is along the bay. However, based on the consensus principle, we might say that this is the poverty that is making those people poor. However, there might be some leaders who might say that this is not poverty. Another leader might say that it is the people who are responsible because they are poor. Because I was born in the same society, I was born

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in the same locality, I come from the same country, and I am not poor anymore. I worked hard in my life. And these people who did not work hard in their lives are poor. However, if we go by the principle of consensus, we will find out that if the behavioral pattern is common with most people out there, we might say that this behavior is externally caused, this behavior is not internally caused. An external cause of this behavior is poverty, not the people who are failing.

Consistency

When we observe people, we look for consistency in their actions. That people behave the same way in many different situations. That someone's behavior is consistent over time. When someone's behavior is consistent over time, it means that they have established this behavior. Because they are consistently behaving in the same fashion.

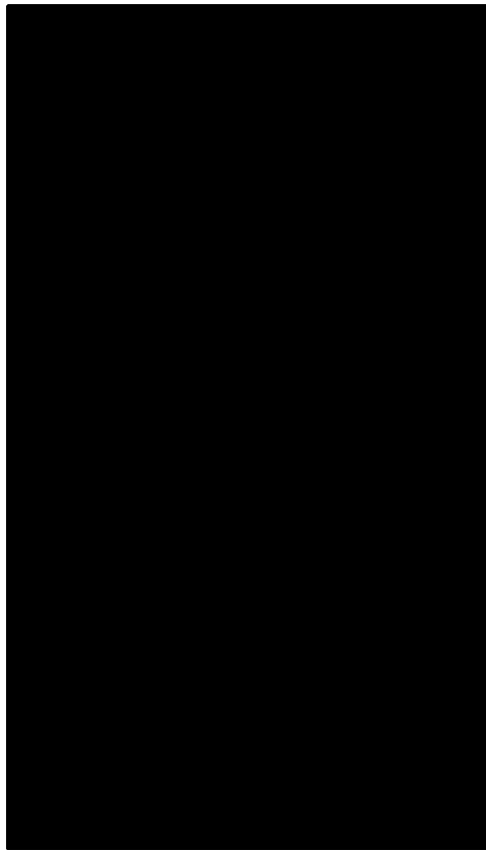
Example: There is a student who fails in every class. He has been failing in every class for the last five years. He hardly passes any exams. And every time the exam comes, this student is upset about it because he knows that he's going to fail it. Who do you think is wrong? Is it the student who's wrong? Or is something happening with the student that's making him do the wrong, which is failing in the exam? Who's wrong here? If we look at the student's behavior, we can say that his behavior is consistent over time. He has been failing for the last three years. And there is something wrong with this chap. So, you say that this chap is poor. You might start commenting upon his intellectual ability, upon his intelligence. You might say so many things to find out the reason inside that person. However, let me tell you that most of the great scientists that have a great name in history were bad students, were the students who were failing consistently. So based upon this perceptual thought, we always say that when a behavior is consistent over some time, then that person is responsible. We do not try to find out the reason in the external environment in the external scenarios.

These are the three important points of attribution theory and individual behavior.

Interpretation

Attribution of Cause

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And the three processes that we discussed are distinctiveness, consensus, and consistency. If the distinctiveness is high, then we would attribute the cause to external factors. If the distinctiveness is low, we will attribute the cause to internal factors. Similarly, if the consensus is very high, we would attribute the cause to external factors. And if consensus is low, we will attribute the cause to internal factors. And the same is the case with consistency. If someone's behavior is consistent over a long period, then we would say that this behavior is caused by external factors. And in other situations, we would say that this behavior is caused by internal factors.

Judgmental Shortcuts

We make certain shortcuts in making judgments. Now, what are those shortcuts that we make when we judge people? These are called judgmental shortcuts. There are the following 5 judgmental shortcuts:

- selectivity
- zoomed similarity. •stereotyping
- halo effect.
- Self-fulfilling prophecy.

Studying these judgmental shortcuts is very, very interesting in the sense that we try, we understand that most of the time that we are trying to learn about people, we are trying to explain their behavior, then we are making certain shortcuts that help us understand their people. But these shortcuts are not always right. These shortcuts give us extremely misleading information about the people. And since that misleading information is going into our minds, and we are building up our opinions and attitudes accordingly, chances are that we would negatively react to our environment in a way that is probably not very conducive for our adjustment over there.

In our next lecture, we will talk about all these judgmental shortcuts.

Lesson 30

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Judgmental Shortcuts and Culture

In this lecture, we will continue the last lecture from judgmental shortcuts and will also discuss culture.

There are 5 types of Judgmental Shortcuts.

1) Assumed Similarity

In this assumed similarity shortcut, we try to find out about others through our own perception of ourselves.

Example: When delivering a presentation, it's common to experience nervousness, a feeling shared by most people. During our initial presentations, this nervousness can be particularly intense, often causing physical shaking. However, as we gain experience, confidence typically grows. Similarly, if we observe someone exhibiting physiological signs of anxiety before a presentation, we may assume they are nervous, reflecting a perceived similarity based on our own experiences.

2) Stereotyping

We start perceiving a group of people in a certain way.

Example: If a person is heightened and muscular we think about him in a particular way that they will be good at sports and think about chubby people in other ways. These judgments can be right or wrong.

3) Halo Effect

We pick one characteristic of a person and start thinking that all characteristics of a person will be good.

Example: In an interview by seeing the color of the tie interviewer makes the judgment that the person will be materialistic and bad for the company.

4) Self-Fulfilling Prophecy

When our beliefs or expectations about a person influence their behavior, it's known as a self-fulfilling prophecy. For instance, if you initially perceived your friend as struggling in mathematics, but later believed their skills improved after tutoring, you might seek their assistance. Consequently, they may also feel more confident in their abilities and willingly assist you, aligning their behavior with your beliefs and expressions about them.

5) Selectivity

When assessing others, we tend to prioritize traits or interests that resonate with us personally. For instance, if you enjoy playing golf, you might inquire whether someone shares this interest. If they don't, you may subconsciously view them as less compatible or not meeting your standards.

Culture

The study of culture is a challenging undertaking because its primary focus is the broadest component of social behavior – an entire society.

To comprehensively study society, it's essential to grasp the collective behavioral patterns it comprises. Culture can be dissected into three primary components:

- 1) The component of beliefs
- 2) The component of values
- 3) The component of customs

Belief Component

In the Pakistan movement, the driving force was a belief component that a different country for Muslims. Muslims are those who have a certain set of beliefs and spend their life accordingly.

Belief and Value Relationship: The beliefs and values component of our definition refers to accumulated feelings and priorities that individuals have about things and possessions as well as ideas.

Values Component

Values are culturally acceptable behavioral patterns. Values are those preferences according to which we conduct or monitor our behavior.

Customs Component

Overt modes of behavior that constitute culturally approved or acceptable ways of behaving in specific situations. Customs consist of the everyday routine behavior of the people.

Example: Talking to elders respectfully is our custom.

How Culture is Learnt

Culture is learned by making use of two processes:

1. Enculturation
2. Acculturation

Enculturation

Learning one's own culture is called Enculturation. Learning about our own culture helps us to adjust. The more we learn the more we get appreciated and adjusted.

There are two ways of learning our own culture:

- Formal (parents taught us)

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- Informal (Through literature or media)

Acculturation

The learning of a new or foreign culture is called **Acculturation**. Culture is a dynamic process, and it evolves and changes. Nowadays a lot of information comes through computers, which helps with acculturation, and we learn about other cultures.

Example: From 1857 to 2009 a lot of changes have been made in our culture. For instance, before that, there was no microwave oven but now in 5 minutes, we can cook food. It has changed our life and culture to evolve like this persistently.

In our next lecture we will study Population and its impact on our Culture.

Lesson 31

Population and Culture

- In our last lecture, we discussed that many factors affect culture like; New technologies, Population shifts, Resource Shortages, Wars, Accidents, Natural Disasters, Changing Values, beliefs, and Customs borrowed from other cultures.
- In Today's lecture, we will talk about population in detail.

But before that, we talk about a culture that preserves itself as it is without any change. **Amish People**

They have preserved their values, beliefs, and customs without any change. They believe worldliness can keep them from being close to God and can introduce influence that could be destructive to their communities and to their way of life. They do not permit the use of tractors in their fields. They do not own or operate automobiles. They also do not use electricity.

They wear plain clothes without designs so named as plain people.

But except for them, 99% of the time in the entire world culture is constantly changing and there is one force behind it and that is the force of population. We are experiencing a population explosion in our cultures.

Population and culture

World Population trends

Current trends in world population point out that by 2030 the world population is likely to be 10 billion and by the end of the 21st Century the numbers may rise to 30 billion. Populations in SubSaharan Africa and The Himalayan regions of Asia have already exceeded the immediate capacity of the area to sustain life.

Pakistan Population trends

Pakistan's population is likely to be doubled by 2050.

Our population is increasing because the birth rate is high, and the death rate is less. So, the growth trend increases.

In the above diagrams, life expectancy and density are given for Pakistan's population.

Factors Affecting Earth

1) Environmental Indicators and population

Based on the population more people use more resources, use more gadgets, and as a result carbon dioxide emissions increase. Stats are given below for these trends.

Vehicles are the biggest danger to the environment. More cars mean more carbon dioxide, so it means that our land fertility decreases and intense damage to the earth. The level of raising up crops decreases.

2) Resource depletion and environmental deterioration

Natural resources start depleting slowly. Resource depletion means the erosion of agricultural soil, Salinization of highly productive irrigated farmland, deforestation, and Lake Acidification (more prevalent in the USA). This includes extensive loss of tropical

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forest, permanent soil degradation (Amazon River Basin) regional water shortages, and deteriorating water quality.

3) Public Policy and the environment

Public policy must Preserve the carrying capacity of the earth, major policy changes coupled with government business and individual actions can do much to alleviate these problems. Important measures need to be taken such as mandating reforestation after cutting, detoxification of chemical by-products before disposal, judicious soil management, energy and material conservation, and industrial and household recycling.

Important facts about Pakistan need to be taken under consideration:

1) Pakistan's urban population to equal rural by 2030: UNFPA

Pakistan's urban population is likely to equal its rural population by 2030, according to a report titled 'Life in the City: Pakistan in Focus', released by the United Nations Population Fund (UNFPA) here on Wednesday.

2) Rural-urban migration (females)

According to the report on Pakistan, the proportion of females is lowest in rural-to-urban migration and highest in rural-to-rural migration. The same pattern has been observed in India. In the ruralurban stream, the share of females is 51 percent in Pakistan. A relatively large fraction of ruralurban migrants' cross provincial boundaries. The perception that "the urban migrant is invariably a male" is incorrect; females make up a considerable proportion of migrants.

3) Slum Population

At least one in every three city dwellers in Pakistan lives in a slum. Many migrants, who move to cities to find jobs and have a better life, may not find jobs in the formal sector or any kind of decent shelter with a minimum of basic amenities. The informal sector employs most migrants, and they gravitate to squatter colonies where they build some kind of shelter for themselves. As a result, slums and marginal human settlements have spread in most urban localities, particularly in urban agglomerations.

4) Increase in Urban Population

The report also shows the share of the urban population increased from 17.4 percent in 1951 to 32.5 percent in 1998. The estimated data for 2005 shows the level of urbanization as 35 percent. The level of urbanization in Pakistan is the highest in South Asia.

More than 60 percent of the population of urban Sindh lives in Karachi and this concentration has increased over time. Approximately three-quarters of the total urban population of Sindh is concentrated in just three urban centers: Karachi, Hyderabad, and Sukkur.

In Punjab, 22 percent of the urban population lives in Lahore, and half of the total provincial urban population lives in five large cities.

Peshawar has a population of approximately one million without counting the Afghan refugees, which is 33 percent of the urban provincial population. The share of Quetta in the total urban Baluchistan population was 37 percent.

More than half of the total urban population of Pakistan lived in 2005 in eight urban cities: Karachi, Lahore, Faisalabad, Rawalpindi, Multan, Hyderabad, Gujranwala, and Peshawar. Between 2000 and 2005, these cities grew at the rate of around 3 percent per annum, and it's projected that this growth rate will continue for the next eight to nine years.

By 2015 it was estimated that the population of Karachi will exceed 15 million, while Lahore and Faisalabad will cross eight million and three million respectively.

This process is called **"Urban Blight"** The population is very Fast moving from rural areas to urban areas.

In the last part of the video lecture lesson 31, interviews with a few Officials from the society have also been conducted for your better understanding in the light of Pakistan and population.

Lesson 32

ImpactOfEnvironmentonItsIncumbents

In this chapter following topics will be discussed:

- Firstly, we'll delve into the relationship between human population growth and the environment.
- Nature and Environmental Changes in Rural and Urban Areas
- Urbanization
- The city as an unnatural habitat
- Crowding
- Rural Areas
- Impact of Environmental Changes on Industrial and Geographical Development
 - Catastrophes and Human Adjustment

Human Population Growth and Environment

The human population of 6000 B.C. was estimated to have been about 5 million people. By A.D 1650 our numbers reached 500 million and in the succeeding two hundred years we had doubled our numbers to 1 billion. In 80 years, by 1930, we had doubled them again.

The current rate of population doubling is approximately 35 years (Ehrlich, 1968); that is, at current growth rates, we can expect the number of human beings inhabiting the earth to double every 35 years and to quadruple within the expected life span of any given individual. **Growth rate = Total Births – Total Deaths**

Put differently, the world's population of humans is currently growing (total births minus total deaths) by an average of over 100,000 people per day (that's approximately one new city of Chicago each month). By the year 2030, barring unforeseen catastrophes, the number of humans vying for space on our planet will exceed 10 billion. The problem is humans do not scatter but make clusters and live in the same place rather than scattering on Earth evenly.

People move toward **urbanization** as they move towards cities more. So, population growth follows the trend of Urbanization. The number of cities with populations over 100,000 has quadrupled in the last 20 years and is expected to quadruple again in the next 20 years.

URBANIZATION

As noted in the chapter's introduction, most of our population lives in or near large cities. Thus, it is not surprising that considerable effort has been expended toward describing and understanding the experience of urban life.

Pakistan Situation

Pakistan's urban population is likely to equal its rural population by 2030.

LifeintheCity:PakistaninFocus',UnitedNationsPopulationFund(UNFPA)

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Urbanization in Pakistan

Sindh: The share of the urban population increased from 17.4 percent in 1951 to 32.5 percent in 1998. The estimated data for 2005 shows the level of urbanization as 35 percent. The level of urbanization in Pakistan is the highest in South Asia. Sindh is the most urbanized province with 49 percent of the population living in urban areas.

NWFP: is the least urbanized province with only 17 percent of its population living in urban areas (according to the 1998 census).

The urban population in **Punjab and Baluchistan** in 1998 was 31 and 23 percent respectively. The urban population in Baluchistan and Islamabad has been increasing at higher rates of 5.1 and 5.8 percent respectively.

Urban Centers of Pakistan: More than 60 percent of the population of urban **Sindh** lives in Karachi and this concentration has increased over time. Approximately three-quarters of the total urban population of Sindh is concentrated in just three urban centers: Karachi, Sukkur, and Hyderabad. In **Punjab**, 22 percent of the urban population lives in Lahore, and half of the total provincial urban population lives in five large cities.

Peshawar has a population of approximately one million without counting the Afghan refugees, which is 33 percent of the urban provincial population. The share of **Quetta** in the total urban Baluchistan population was 37 percent.

Urban Population Concentration

More than half of the total urban population of Pakistan lived in 2005 in eight urban agglomerations: Karachi, Lahore, Faisalabad, Rawalpindi, Multan, and Hyderabad. Gujranwala and Peshawar between 2000 and 2005, these cities grew at the rate of around 3 percent per annum, and it's projected that this growth rate will continue for the next eight to nine years.

Urbanization Growth Rate

By 2015 it is estimated that the population of Karachi will exceed 15 million, while Lahore and Faisalabad will cross eight million and three million respectively. According to the UNFPA global report, more than half of the world population, around 3.3 billion, will be living in urban areas by 2008 and the number will swell to around five billion by 2030.

Country representative of UNFPA, Dr France Donnay said that the growth of cities would be the single largest influence on development in the 21st century. But little was being done to maximize the benefits of urban growth or reduce its harmful consequences. "Between 2000 and 2030, Asia's urban population is to increase from 1.36 billion to 2.64 billion and Africa's from 294 million to 742 million".

Urbanization results in the formation of slums and affects cities' infrastructures and resources.

Slum Formation

At least one in every three city dwellers in Pakistan lives in a slum. Any migrants who move to cities to find jobs in the formal sector or any kind of decent shelter with a

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minimum of basic amenities. The informal sector employs most migrants, and they gravitate to squatter colonies where they build some kind of shelter for themselves. As a result, slums and marginal human settlements have spread in most urban localities, particularly in urban agglomerations.

"Katchi Abaadi" or Slum Statistics

In Pakistan, the urban population living in "*katchiabadis*" varies between 35 and 50 percent. The growth of these informal settlements in the two megacities, Karachi, and Lahore, has particularly been massive. In Karachi, these settlements increased from 212 in 1958 to more than 500. In Lahore, there are more than 300 katchi abadis in Faisalabad, at least 40 percent of the population lives in these abadis.

Effects of Urbanization

There can be the following effects of urbanization:

- Physical Effects
- Social Effects
- Psychological Effects

When urbanization increases, people start moving to the suburbs of the city and all of this remains unplanned, and this urbanization keeps moving like a vicious cycle. As a result, following **physical effects** occur:

Psychological Effects

In cities, people become busy and don't find time to become friends with others, start feeling lonely and problems increase.

Anonymity increases as people don't know each other and hence loneliness increases as well as the crime rate increases, and privacy is compromised here.

Lesson 33

Effects of Urbanization

In this lecture, we will continue the psychological and physical effects of Urbanization.

Physical and Psychological Effects of Urbanization

Then there are problems related to crowding. **Crowding** means that more people are sharing a certain area. That means people might be closer to each other than this is comfortable for them. An example could be one gentleman who has been living in probably some rural area and moves to Karachi and starts living in an apartment building, he finds out that people are very close for comfort. And they are scooped up into small little dingy apartments where everybody is living their lives.

So, these people might find out that their privacy is being confiscated.

Secondly, because of urbanization **pollution** also rises. When pollution rises, then some physical diseases and disorders are also in their eyes.

There would be more food required and the supply chains for such food probably might not be very well in third-world countries.

So, **starvation** might also be one of the negative outcomes of the big cities. With this **drug and alcohol abuse** has been found more in the bigger cities than in the rural communities. And there has been **family disorganization**. It's been said that in Western countries, the institution of families is deteriorating, and families are disintegrating. People living as a family are much fewer than they were used to in the past. This has been seen that as people move into the urban areas, some kind of family disintegration starts taking place because families go too far for the comfort of the people. They are constantly working; they are constantly taking different types of pressures. So, family disintegration becomes one of the major negative outfalls of urban lives. However, it does not happen so frequently.

So, students, these were some of the physical and psychological factors that people suffer from in urban areas.

Problems of Population/Urbanization

There were two main problems:

- One is that the population is growing very fast,
- And second is population is following the trend of urbanization.

Now we will understand these problems under multiple theoretical frameworks.

Theories of Urban Effects

1. Wohlwill's Adaptation Level Theory
2. Milgram's Information Overload Theory
3. Proshansky, Ittleson, Rivlin- Behavior constraint Theory

1) Wohlwill's Adaptation Level Theory

Adaptation level theory means that human beings function best when their level of arousal is on an intermediate scale.

This means that they are not too much aroused, that they are not too under-aroused.

When we apply this theory to city life then we find out that there is a lot of population in

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the city. There are so many entertainment places.

Then so much is going on all the time in the city. A gentleman who comes from a far-off village to a city in Karachi or Lahore might say that this city never sleeps. Even at two o'clock at night, you could find cars on the roads, you could find out the food hangouts are open, and a lot of activity is going on throughout the night. So, apply the adaptation level theory. You will find out that in the cities, people are constantly stimulated. Low-income, constant stimulation will be here. Or if stimulation will be here constantly, that means the arousal level is up. And when the arousal level is constantly up, then what happens?

An arousal level increase means that a lot of hormonal activity is going on. People might bump into a fight-or-flight reaction. So, you can find out the problems like low concentration, and low performance level.

2) Milgram's Information Overload Theory

That means if more information is coming our way, then we can handle it, then our behavior, and we simply start missing out on that information. This means that we are so loaded with information that we don't know how to find a way out of it, how to analyze it, or what to do with this. According to the information load theory, the quantity and rate of stimulation that urbanites are exposed to exceeds their capacity, resulting in information load. So, another way to understand the city would be that concerning information load, you will find a lot of people who complain about information load. Because the city is constantly stimulating them, the city is constantly providing them with something.

3) Proshansky, Ittleson, Rivlin - Behavior constraint Theory

The demands of city life limit an individual's freedom, leading to behavioral constraints that can produce feelings of loss of control, leading to behavioral constraints. For instance, a boy living in the village moved to the city because he must study at a university now. When he comes to the university, he finds out that situations are so complicated, he must work very hard to find his way into it and way out of it. Eventually, he bumps into behavioral constraints, which means that for some time, he would start feeling that he's kind of losing control over the situation. And when people are losing control over the situation, what kind of psychological feeling do they go through? That means they're feeling frustrated. That means they're feeling depressed about the entire thing. Because the entire situation, the entire environment becomes so complicated for them, and it is charging so heavily upon their senses that they find it hard for themselves to kind of control it. And that type of feeling creates frustration, anger, and remorse, dejectedness, depression.

Now, can you correlate these feelings with the incidence of crime somehow? Do you think that a satisfied person who's happy about himself, do you think a person who's calm about things and who has control over the things around himself, has some inclination towards crime, the possibility is very less? Compared to people who are going through extreme psychological feelings, the incidence of crime in such a state of mind might be more prevalent. There's more possibility for the crime to take place when a lot of people are going through such a mental state. So, we can apply this theory also.

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At this point, it is also interesting for you to note that a gentleman, a psychologist whose name is Baker, said cities are overmanned. In cities, there are so many people living together that the city's infrastructure chokes. And when the infrastructure chokes, then people bump into negative behavior patterns. And when people bump into negative behavior patterns, that might be devastating not only for themselves but also for the people around them. These are some of the theories that we can apply to understand the urban effects.

Urban VS Rural

Number one is urban-rural comparisons. Number two is social behavioral effects. Number three is beneficial effects.

Find out a friend of yours or a cousin of yours who lives in a big city and ask him what charges him. Ask him about his routine. You will find out how urban and rural life differs. Urban life seems to be more complicated. People seem always to be on the run. Urban life can be the lowest income in the years of the year. When you get into the villages, you find out that rural life is kind of simple. This is why if you have spent a long time in the city, you probably might point out yourself to be very peaceful in village life. You have different facilities that you are so used to using in the cities that are not available to you in the village. You might bump into a different type of reaction. You might not actually like that place. However, environmental psychology is the subject. This study the effect of the environment on human behavior. In this very comparison, we can find out that the environmental features in urban life and rural life are two distinct and different types of features. And they impact human behavior differently. So, it is very natural that we can always compare urban life with rural life.

And for the start of it, when people compare both lives, they bring up their own points of use into that. They might say that rural life is very good, calm, and serene. And then there might be people who might say that urban life is wonderful because you have all the facilities and entertainment available to you. We start our comparison with the physical and health diseases that are found prevalent in the people of rural areas and the people of urban areas. **1)Physical and Health diseases**

Statistics show that there is probably not uniformly worse physical and ornamental health in cities compared to rural life. There is a higher rate of respiratory diseases in the cities. That is understandable because in cities there is a lot of pollution. The minimal differences in the incidence of stress-related illnesses such as heart disease and hypertension. This research was carried out in old research (Hay & Wantman, 1968).

Then it is seen that alcoholism and drug addiction are more common in cities than in rural areas. It has also been found that the incidence of psychosis is higher in urban areas than in rural areas. This research was conducted by Doran Mann and Doran Mann. So, students, when we compare urban and rural areas concerning the health and disease factor we find out that there is not much significant difference over here. Certain diseases are more specific to city life and there might be certain diseases that could be more specific to urban life. **2)Social and behavioral effects**

Differences between urban and rural settings can be observed concerning two important psychological factors. Number one is **affiliation** and number two are **pro-social and anti-social behavioral patterns**.

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Affiliation

When it comes down to affiliation we find out that urbanites are less affiliative towards strangers than are people in the rural areas. This research was conducted by Macaulay, Coleman, and Defuso in 1977, then Milgram and Newmann, and Macaulay once again studied it in the same period. They also found out that urbanites tend to avoid eye contact with strangers and are less likely to reciprocate friendly gestures than rural dwellers. That means people living in the cities are very bad because they have a lesser affiliation towards other people than the people living in the rural areas. Well, these are the statistical findings. This is some of the research that has been conducted and these are two different behavioral patterns. It does not mean that people living in the city are bad or people living in the rural areas are good or bad either. It means that both people are exposed to two different types of environments and since they must survive that environment, different behavioral tactics work in one place and different behavioral tactics work only in the other place. In rural areas, few people know each other, and usually, the threat to the community or the fabric of the community is very, very strongly woven. In the cities, the threat or the fabric of the community is not very strongly woven because there are people from all over the country who are living in the cities. As we discussed in our last lecture, people also might be suffering from the feelings of anonymity and people might be feeling that there are a lot of strangers living in this place, they might have been using this type of behavioral pattern as a protective measure to survive in the city. This is exactly what these two researchers found out.

Pro-social Behavior

It has been observed that urbanites are less likely to help a stranger. This research was conducted by Galphand, Hartman Welder, and Page, then Kurt and Kerr as well as Melkram. Urbanites are less likely to help a stranger. What does it mean? Urbanites are selfish people or people living in urban areas are not good ones. It could mean so many different things. But it has been found that since in the city, a lot of people are living together and people are from different areas, the level of anonymity is high. The level of feeling stranger is kind of very high. And then simultaneously in the city, the stimulation level is also very high. People are constantly aroused. There could be the application of behavioral constraint theory. People tend to lose control over the scenario. And since they are already losing control over the scenario, they might not use extra energy for the strangers. If you apply the stimulus adaptation level theory, you might find out that so many stimulants are available to the people in the city that little energy is left for the stranger. Then of course in the city a lot of trainers are living and then people are from so many different places, you don't know about them. You probably would not be able to trust them so easily. So, this is one type of behavioral pattern that is found concerning pro-social behavior.

Anti-social behavior

Anti-social behavior is the exact opposite of pro-social behavior. Pro-social means that you are pro-people. You are towards the people. You affiliate with them. You connect with them. You relate to them. And what is anti-social behavior? Anti-social behavior means that you are against people. You don't want to connect with them. Or you want to take advantage of them. So, when we talk about anti-social behavior, you will start a

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behavioral pattern that will be used in your crime.

Statistics: Clear differences have been reported between herbal and ruined areas in the incidence of crime. This research was conducted by a courtsman, and Levy, fisher, and Zimbardo. They have found out that in urban areas, the incidence of crime has been quite high. And in the rural areas, the incidence of crime has been quite low. Now what can be the reasons? The factor of anonymity might play a role in the crime. Then if you apply the adaptation level theory, we understand that people are constantly stimulated. Their arousal level is constantly high or constantly low. They are not performing very well. And when they are not performing very well, probably their efforts towards achieving certain objectives are also not very successful. That leads to frustration and dejection that might lead to the incidence of crime also. Then if you apply the behavior constraint theory, loss of control over things. A young boy comes to the city and wants to be a rich man. He probably does not find good opportunities. He gets so frustrated and upset. And he thinks that I don't know anyone here. And this is why so many different people are living. It might decide to bump into a crime. So, in city life, it has been found that the incidence of crime has been higher than in rural areas. And there are some understandable facts about that also.

There is everything negative about the cities. Is that so? Now do not mean to say that. There is, it is not like that that everything is negative about the city. The city life is kind of different. Otherwise, if you ask most of the people, they will not say that they want to live in the cities or they want to live in the urban centers.

Beneficial effects of city: There is an interesting fact to note, which was researched by Malton and Hargrove says the city provides you with all the infrastructure where you could become a more sophisticated person in your career. There are so many different options available in the city, the business is going on. In the city, all the opportunities are available to you. So that is why cities are places where people find more opportunities. When they find more opportunities, they can express their inner potential and their inner talent. And once they can express their inner potential and talent, probably that might be a satisfaction for them. And of course, people make more money in the cities. So, they say that rural scenes are substantially more likely than urban scenes to be described as pleasant turns. Rural inhabitants are more likely to be perceived as friendly. This was research that was conducted by Malton and Hargrove in 1987. Looks like this research once again is telling us that cities are bad places because a lot of people find out that the rural scenes are more beautiful. They find out that the rural people, rural inhabitants are more friendly than the city inhabitants. How do you understand this research? This research is simply a comparison that compares people from rural or urban areas concerning behavioral patterns. So, as we discussed when you look at the city from the positive point of view, we find out that much of the research starts with a biased and pessimistic assumption that cities are bad. Most of the research has probably been carried out with a certain bias in the mind of the researcher that the city is something bad. It's a bad place to live. And this fact was researched by Friedman and that's also quite old, way back in 1975.

According to a Gallup opinion index, most of the respondents indicated they preferred living in cities. So, there are some positives also attached to the cities. Life in the city makes an individual more versatile and adaptable and gives a broader perspective on life than is afforded in rural life. So, these are some of the ways rural areas and urban

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areas can be compared.

Crowding

Considering the above-mentioned topics, we will study crowding in detail under the following topics:

- Animal Studies
- Human Correlational/survey studies
- Human Experimental Studies
- Social Affective Responses
- Crowding in everyday setting

Crowding means a lot of people are living in little space. When more people are living in little space, then that means their privacy has been hampered. That means they are probably not feeling comfortable with each other. So crowding is an artificial state where people are too close for comfort. Crowding means that you are too close to comfort with the people. This means the density of the population is ticking and people are so close for comfort with each other that they are not feeling comfortable or peaceful with each other anymore.

Animal Studies

Naturally, we can use many ways to do our studies or our research. However, it has been found that whenever we are doing psychological research, since it involves human beings, sometimes conducting experiments becomes very, very dangerous. So, the experiments are conducted on animals. Huge caution needs to be taken from animal study skills that are generalized to human beings. Secondly, animal studies have a direction to do this. It gives us a strong hypothesis that is almost there to be proven that this is the way human beings can also actually act. Animal studies are very difficult to understand. We can use them in 2 lines to understand the research.

Naturally occurring population cycles: This is how we can naturally do population cycles.

Experimentally controlled population cycles: This means the population is experimentally controlled. There are certain conditions. There are experimentally controlled species of a certain animal. There is another group that is not being applied to any kind of independent variable, which means that no changes have been imposed upon this group's growth of population. They let it happen very, very naturally.

1)March to the Sea Study (Natural Population Cycle Study)

This research was conducted by Dubos in 1965. Whenever we want to understand what happens concerning a population of human beings, this becomes very, very interesting research to code. You can see an animal that closely resembles a rat. This is a Norwegian Lemming. Norwegian Lemmings are small rodents resembling the field mouse but having short tails and covered feet. They live primarily in the Scandinavian Mountain regions. Lemmings can be used by the natives that about every 3 or 4 years they appear to migrate or march to the sea with many of them drowning as a result. This is very, very strange behavior. Lemmings are growing up, and up to a certain extent they grow, and then they certainly decide to commit suicide. They simply move towards march towards the sea and drown themselves up so that they can balance their

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population. So, what exactly is this? Do you think Lemmings committed suicide? Because they are too many in number and some of the people, some of the Lemmings decide to sacrifice themselves. What is happening over here?

When we do a closer observation, we find the Lemmings when they are marching toward the sea, this moment is not very orderly. This is rather full of frenzy. So, what is going on? Is it suicide or an accident? They find out that certainly Lemming's bump into a frenzied activity or helter-skelter on the Lemmings are just madly running towards the sea or in some Lemmings simply drown themselves in the sea. So closer observation reveals that this is not an organized thing. This is a disorganized thing. This is some kind of madness that the Lemmings are experiencing, or the Lemmings are going through. Lemmings are prolific reproducers. Every three or four years they reach considerable numbers. This increased population density acts to influence brain and adrenal functioning which in turn is overly manifested in non-directed activity. They found out that when Lemmings reach a certain level of population then in Lemmings a very heightened level of adrenal activity starts taking place. Adrenal activity is very closely related to frenzied behavior. So biological pre-programming means that they simply bump into frenzied because of the adrenal gland glandular functioning and then in that frenzied, they don't know where they are going, and they don't know what they are doing, and they simply commit suicide, or they simply die and drown themselves into the sea.

The question is do human beings do the same thing? Human beings are taken as lemmings and the population becomes more and they are jumping into the sea. Are there any Lemmings humans? Does the population create some impact on us? Well, we can give you another interesting study.

That will take the story forward. And this interesting story is called Dear Die Off on James Island.

Dear Die, Off on James Island

This research was conducted by Christian Figer and Davis in 1960. In 1916 4 or 5 deer were released on James Island in Chesapeake Bay off the coast of Maryland. By 1955 this small herd had grown to 280-300 deer. In 1958 half of the herd died. This was something very alarming what happened with all these deer? How come they die? They had crossed their population level up to a certain extent and then they simply started dying. What happened to them? They found out that the herd stabilized on the threshold of 80 members. From 1958 onwards they maintained the level of threshold of 80 members. So, what was happening with these deer? A closer examination was conducted by Christian Scrope and this is what they found. Scientists John Christian Fliger and Dave Devas studied 18 after surviving deer. They performed a detailed histological examination of the deer's adrenal glands, thymus, spleen, thyroid gland, kidney, liver, heart, lungs, and other tissues. The care cast was found to be in excellent shape throughout. The only abnormal finding was an increase in the size of the adrenal glands. In most severe cases, adrenal glands were found to be 10 times the size of the base rate samples. They found out the deer were in excellent shape. There was nothing wrong. Their heart spleen, they were, everything was functioning fine, and they had the right amount of fat in their body which shows that they were healthy deer, but they found only one abnormality, and that abnormality was died der had 10 times bigger

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adrenal glands. The **adrenal gland** is the gland that controls the thinking activity that controls the behavioral activity of an animal. So, if the adrenal gland is 10 times thicker than the normal adrenal gland which is found in most deer, then what is the reason for that? You would find out that the Christian's crew came up to be forming a very interesting conclusion. They said they determined that while they were in good nutritional condition, their physiology indicated hyperstimulation from the stress of crowding and attributed the many deaths to this. You remember Wahlbl's theory, adaptation level theory, that when people perform best at the normal level of arousal, at a medium level of arousal. In this case, the deer is overstimulated, which means he is not performing at the normal level. Deer are overstimulated because of the overcrowding. They are too close to each other probably. And that creates stress. That stress leads to these biological deformities and eventually, the deer dies.

We find out one thing about the animals, that the animal's instinctually controlled behavior is usually very, very huge. However, when we come down to human beings, we find out that there is reason and logic. That is why human beings are more adaptable, they adapt more and more. They change more and more. They just want a human being that is surviving. So, however, these studies point out one thing that more population, if there is more population, that is some kind of discomfort.

How are we going to deal with this? We can talk about this in the next lecture.

Lesson 34

Problems Related To Crowding

Crowding-related issues are significant topics of investigation within the field of environmental psychology. Research on animal behavior in crowded conditions provides valuable insights into the impacts of population density on various aspects of physiological and behavioral functioning. **Animal Studies of Crowding**

Animal studies concerning crowding can be categorized into two primary streams of inquiry. Firstly, investigations focus on naturally occurring population cycles, while secondly, researchers experimentally control population concentrations.

a. Naturally Occurring Population Cycles

Examples of it include studies such as the lemmings' phenomenon of migrating towards the sea, as observed by Dubos (1965), and the die-off of (Sika) deer on James Island, as documented by Christian, Flyger, and Davis (1960).

The migration of Norwegian lemmings, small rodents resembling field mice, every three to four years, often culminates in many of them drowning. While Scandinavian folklore attributes this behavior to a deliberate suicidal tendency ingrained in their biology, closer examination reveals it as a frenzied, rather than orderly, movement. This phenomenon suggests that increased population density influences brain and adrenal functioning, leading to non-directed activity.

Similarly, Christian, Flyger, and Davis (1960) noted abnormal adrenal functioning in a herd of deer experiencing increased population density. Their observations on the herd's health, particularly the enlargement of adrenal glands, indicated heightened stress levels due to crowding as a probable cause for the observed physiological changes.

b. Experimentally Control Population Concentrations

In contrast, controlled experiments, such as those conducted by Calhoun (1962) with Norway rats, provide further insights into the consequences of overcrowding. In this study, Calhoun provided Norway rats with an environment characterized by abundant resources, including unrestricted access to food, water, and nesting materials. Notably, the only limiting factor imposed was space, as the physical environment remained constant throughout the study. Over time, the rat populations expanded, leading to a phenomenon where dominant individuals claimed more spacious territories, while others were confined to increasingly crowded areas within the confined space. This confined area was termed a "behavioral sink" by Calhoun.

Within the behavioral sink, observable behavioral and physiological abnormalities emerged among the rat population. For instance, nesting behaviors among females became incomplete or poorly executed, resulting in neglected pups and disrupted nursing behaviors. Additionally, juvenile males exhibited pansexualism and instances of cannibalism, while varying levels of aggression and withdrawal behaviors were observed among different individuals. Furthermore, increased

rates of infant mortality and aborted pregnancies were noted. Physiological abnormalities included the development of tumors in mammary and sex glands among females, as well as anomalies in the kidneys, livers, and adrenal glands in both sexes.

Calhoun's research prompted inquiries into the potential association between population density and the prevalence of pathological conditions in humans. The pathologies identified by Calhoun, including morbidity, mortality, fertility issues, ineffective parenting, and psychiatric disturbances, or their related manifestations such as infectious diseases, juvenile delinquency, and adult criminality, have been tentatively linked to escalating human population densities, albeit with varying degrees of success. This body of work underscores the complex interplay between environmental factors and behavioral and physiological outcomes, providing valuable insights into the dynamics of crowded environments and their potential impacts on human well-being.

Additional effects of increased population concentrations have been reported subsequent to Calhoun's seminal work. For example, it has been shown that population density leads to the production of fewer sperm by males, to later onset of estrus, less frequent and shorter duration estrus, and consequently less frequent births and smaller litters (Crow & Mirsikowa, 1931; Snyder, 1966; 1968) among females. There is also evidence for an increase in alcohol consumption and performance decrements on complex tasks (Goeckner, Greenough, & Maier, 1974) for rats raised under high-density conditions.

Challenges in Extrapolating Findings to Human Behavior

Similar effects have been observed across a diverse range of species including swine, chickens, cattle, elephants, and various rodents (Greenburg, 1969; Hafez, 1962; Marsden, 1972; Theissen, 1966). However, caution is warranted when extrapolating these findings to humans:

- Firstly, the behavior of non-human animals is often more biologically determined and less influenced by learning and cultural factors (Swanson, 1973).
- Secondly, humans typically have opportunities for temporary relief from high-density environments, unlike animals studied under controlled conditions (Evans, 1979).
- Lastly, humans have demonstrated superior capacities for adaptation and adjustment compared to other animals.

Value of Animal Research in Understanding Human Behavior

Despite the difficulties in using findings from studying animals to understand humans, this research is still really important for learning about human behavior. Animals can help us come up with ideas about how people might behave, especially when it's not ethical to test on humans directly. For example, if we want to see how living in crowded places affects future generations, it's better to use animals that have shorter

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reproductive cycles than to use humans. Also, with new technology, we can monitor animals' behaviors and bodies without hurting them, which helps us avoid ethical problems we might face with human studies. But it's also important to note that we do research with humans too, and there's a lot more to learn in this area.

Lesson 35

Impact of Population Concentration among Humans

Population concentration among humans has been a subject of significant interest in environmental psychology, particularly concerning its effects on crime, mental illness, and social disorganization. This section explores early research and contemporary perspectives on the impact of population density on human behavior.

Historical Perspectives

Early investigations focused on the correlation between population concentration and various social indices, such as crime rates and mental health indicators. Studies have consistently shown higher rates of violent crimes and worsening mental health conditions in densely populated urban areas compared to rural regions.

Allure and Pessimism

The idea that increased population concentration could be responsible for these effects presents both pessimistic and alluring implications. While acknowledging concerning trends, researchers are intrigued by the possibility of understanding human behavior through mechanisms similar to those affecting other animals.

Defining Density

Investigators have utilized various measures of population concentration, often interchangeably referring to them as crowding or density. However, consensus on the appropriate measure remains elusive.

Schmitt's Measures

Schmitt (1957), for example, assessed the relationship between five semi-independent measures of density and adult crime and juvenile delinquency.

They used the following five measures

- Average household size
- Proportion of married couples without their own home
- Proportion of dwelling units in structures with four or more living units
- Population per acre
- Percentage of occupied dwellings with 1.51 or more persons per room

Drawback of the above measures is that only the last two showed strong positive relationships with delinquency and adult crime.

Inside vs. Outside Density

In subsequent work Schmitt (1966) has distinguished between inside density and outside density. That is, Schmitt noted the difference between the number of people per unit of living space and the number of people in the larger community (e.g., persons per acre). The importance of these contrasts becomes apparent when the following possibilities are considered: low inside-low outside (suburbs); high inside-high outside (urban ghettos, barrios, and others); low inside-high outside (luxury inner city apartments and condominiums); high inside-low outside (rural farm areas).

Drawback of Relying solely on inside density would assume uniform effects across different environments, while considering outside density provides more nuanced

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predictions. These insights underscore the importance of precisely defining the term density.

A second challenge highlighted by the earlier considerations pertains to the differences in socioeconomic, ethnic, and social structure variables among the four areas under study. These variables might independently contribute to degenerative or pathological effects, but their covariation with density can lead to misattribution of causality.

Galle, Gove, and McPherson Study

Recent research has aimed to address this issue by adopting more explicit measures of density and carefully accounting for confounding variables related to social structure. For instance, in a study conducted by Galle, Gove, and McPherson (1972), correlations were observed between persons per acre and public assistance rate, juvenile delinquency, and admission rates to mental hospitals. However, these correlations disappeared when controlling for social class and ethnicity, suggesting that density might not be directly related to social disorganization once these factors are considered. Given these findings, Galie's group undertook a reanalysis of their data. In this subsequent analysis, density was defined differentially, encompassing factors such as:

- the number of residential structures per acre,
- the number of living units per structure, the number of residents per living unit,
- and the number of persons per room.

Utilizing multiple correlation techniques, they demonstrated that these refined measures of density significantly correlated with various pathologies, even after controlling for class and ethnicity. Their findings indicated that the number of persons per room was the most significant contributor to various pathologies, except for admissions to mental hospitals. Additionally, factors such as the percentage of people living alone and the number of persons per living unit emerged as strong predictors of certain pathologies.

Drawback of the above studies was pointed out by Kellett (1989), whom has questioned the adequacy of density as a concept for investigating the relationship between health and housing, suggesting that further scrutiny is warranted.

Lesson 36

The Distinction between Density and Crowding

Recent research by scholars such as Altman (1975), Loo (1973), Proshansky et. al (1970), and Stokols (1972a, 1972b, 1976) has highlighted the crucial difference between physical density and psychological crowding. Stokols (1972a) notes the tendency to interchangeably use the terms density and crowding in human research, emphasizing the need for distinct definitions. While density refers to physical limitations of space, crowding represents an experiential state influenced by personal and social factors in addition to spatial constraints. Thus, Stokols proposes that density is a necessary but not sufficient condition for crowding. As Stokols views it, the potential inconveniences of limited space such as the restriction of movement, the preclusion of privacy, and other disadvantages of space limitation must reach some degree of saliency and be viewed as aversive before people experience crowding.

Diverse Definitions of Crowding

Various scholars, including Desor (1972), Zlutnick and Altman (1972), and Van Staden (1984), have proposed different definitions of crowding. For example, Desor (1972) has defined crowding in terms of excessive stimulation from social sources; Zlutnick and Altman (1972) have defined crowding in terms of an individual's inability to adequately control interactions with others; Van Staden (1984) emphasized perception of large numbers of people, spatial restriction, and an aversive experience of the situation as relevant components of the crowding construct. In this regard crowding may be thought of as a psychological concept referring to a subjective experience, which may or may not be adequately reflected by population density measures such as the amount of physical space per person or number of people per unit of living space. In short, there seems to be general agreement that high density is seldom a sufficient and probably never a necessary condition for the experience of crowding.

Complexity of Density and Crowding

Density, particularly areal density, encompasses multiple components such as residential average, structures per area, dwellings per structure, rooms per dwelling, and persons per room. This complexity highlights the variation in neighborhoods with similar population densities but distinct land use patterns, affecting the experience of density differently. Operationalizing crowding proves challenging, with current approaches relying on correlational/survey methods. However, limitations in controlling extraneous variables and definitively linking density to pathology hinder conclusive findings.

Laboratory Studies for Enhanced Understanding

To address these challenges, environmental psychologists have turned to laboratory studies to manipulate physical and social parameters of crowding more effectively. For instance, Arkkelin et al. (1982) developed simulation techniques allowing independent manipulation of density components and found that interpersonal distance interacts with social variables to influence subjective discomfort more significantly than the absolute number of people present. Their research suggests that crowding is characterized by heightened arousal, displeasure, and a sense of loss of control. Subsequent sections will review additional studies exploring density manipulation and its impact on affective and social-behavioral responses.

Crowding & Density Human Experimental Studies

Conducting carefully controlled experimental studies of the effects of population density on human behavior involves many difficulties. Nonetheless, a number of studies have been conducted and reported. Some research has utilized rather interesting, but naturally occurring, high-density situations like prisons, commuter trains, playgrounds, and shopping centers. Others have purposely manipulated either numbers of persons within a setting while holding setting size constant (i.e., manipulated social density) or have maintained the same number of respondents while manipulating the setting size available to them (i.e., manipulated spatial density; see, for example, Loo, 1972; McGrew, 1970; Pedersen, 1983; Saegert, 1973, 1974). In keeping with the previous chapters, we will look at what is known about the effects of density on physiological processes, on task performance, and on such social affective behaviors as attraction, altruism, and aggression.

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Physiological Reactions

One of the first researchers to report on the physiological effects of density was D'Atri (1975). He observed the blood pressure levels of inmates in a prison who were confined to either single or double occupancy cells and found those inmates in the latter cells to exhibit higher levels. Paulus, McCain, and Cox (1978) have also found blood pressure to be positively related to increased density, and McCain, Cox, and Paulus (1976) have reported that inmates in high-occupancy dormitory settings complain more of illness than do those in lower-density cell blocks. This finding may occur as a result of real illness brought on by physiological disturbances or by a desire on the part of the inmate to be taken from the high-density dormitory to the lower-density infirmary. In either case, the differential density seems to be the reason for the discrepancy in the number of complaints.

Task Performance

Early research on the effects of density on task performance utilized relatively simple tasks like psychomotor tasks, problem solving, or anagram solving in laboratory contexts where subjects either knew or could easily discern the interests and hypotheses of the experimenter. Saegert, Mackintosh, and West (1975) found support in an interesting and provocative field study for the notion of increased density leading to performance decrements.

Saegert and her colleagues tested subjects in a socially dense department store and in a busy railway terminal. Among the tasks subjects were asked to perform was to provide a cognitive map (see Chapter 4) of their environment. The researchers concluded that tasks which involved knowledge and/or manipulation of their environments were impeded by increased density. The fact that increasing density is accompanied by a corresponding decrease in the clarity of one's mental image and knowledge of the immediate environment may also explain the findings of Glassman, Burkhart, Grant, and Vallery (1978) and Karlin, Rosen, and Epstein (1979). These researchers showed that social density was related to grades attained in two separate university settings. Additionally, Karlin et al. (1979) showed that once conditions of high social density were removed (i.e., students were reassigned to less crowded quarters) grades improved significantly. Thus, increased density may overload one's information-processing ability, resulting in impaired performance on tasks that require higher-level cognitive skills.

Social-Affective Responses

It is not difficult for any of us to recall situations of high density that have made us feel uncomfortable. Crowded elevators, bargain basement sales, the first day of classes in a large lecture hall, bus depots and airport terminals where we are hurrying to make connections are but a few. Without much difficulty we could probably also recall occasions when this discomfort has led to feelings of tenseness, anxiety, or even anger. On the other hand, parties, rock concerts, and crowds at athletic contests also represent situations of high density that may have evoked very different, perhaps even very

positive, feelings. Hence, our feelings are obviously related not only to density but to the circumstances under which density occurs.

A number of studies have investigated the effects of density on social behaviors such as attraction, altruism, and aggression. Let us examine these findings.

- **Attraction**

The first laboratory study to show that interpersonal attraction (liking) is reduced under conditions of high social density was conducted by William Griffitt and Russell Veitch (1971) at Kansas State University. Subjects in that experiment were exposed to one of two conditions of social density and one of two levels of ambient temperature. After being given time to become inured to these conditions, participants were asked to make evaluations of strangers and to indicate their degree of probable liking for them. Subjects who were exposed to the high-density conditions gave significantly more negative evaluations and expressed a lower degree of probable liking than did those exposed to low-density conditions.

- **Aggression**

It might be expected that if differential density leads to diverse emotional responses, then to the extent that aggression is related to mood, aggression might be expected to vary as a function of density. Density does not automatically lead to a specific emotional response. For this reason, and following the above logic, aggressive responses would be expected only to the extent that density elicits negatively toned affective responses. This elicitation results not only from the density of the environment but by the personal and social conditions of the setting. For example, Rohe and Patterson (1974) have found that high density leads to aggression in children when the play situation does not provide ample toys for each. As long as there are plenty of toys to go around, increases in density did not affect children's aggressive behavior. Thus, it is likely that density, like temperature, acts only to moderate aggressive responses and is not the direct cause of them.

- **Altruism**

Altruism studies suggest that density doesn't directly impact helping behavior, but rather influences it indirectly based on other factors. Factors like environmental affect, fear for personal safety, and positive mood states play roles in determining helping behavior. Discrepancies in helping behavior between rural and urban areas may be due to differences in attentional processes or varying lifestyles affecting available time for assistance.

Crowding In Everyday Settings

In the field of environmental psychology, researchers have explored how living spaces affect people's well-being. However, their findings have not always been consistent.

Residence and Density

Researchers studying environmental psychology have explored how living spaces affect people's well-being, particularly in relation to density. Sweaney et al. (1986) found that as the number of people in a living space increased, children tended to feel more crowded. On the other hand, Duckitt (1983) discovered that dissatisfaction with living

spaces could occur in both high-density and low-density environments. In a related study, Gabe & Williams (1986) noted that students living in rooms meant for three people but occupied by two had lower self-esteem than those in rooms designed for two people. However, Ronchi & Sparacino's findings did not show significant differences in how people rated the pleasantness or arousal levels of different living densities.

Why Inconsistent Results?

The inconsistencies in research findings could be attributed to several factors. One major factor is the differences in how researchers conducted their studies and defined terms like "density" and "crowding." These variations in methodology may have led to differing conclusions about the relationship between living density and psychological experiences. Additionally, it's possible that there isn't a consistent relationship between living density and psychological responses, highlighting the complexity of this area of study.

Social Variables and Density

Some studies have suggested a link between living density and suicide rates, but this relationship may be influenced by social and cultural factors. For instance, perceptions of population density vary between different cultures, such as the differences observed between the United States and China. People from different cultural backgrounds may also exhibit varying levels of tolerance for high-density living. Gillis, Richard, and Hagan (1986) noted that Asians tended to be more tolerant of high-density living, while British individuals were less adaptable, and Southern Europeans fell somewhere in between these extremes.

Cultural Adaptation to Density

Long-term exposure to different population densities could contribute to varying levels of adaptation across cultures. Gillis, Richard, and Hagan's (1986) study highlighted these differences, with Asians showing greater adaptability to high-density living compared to their British counterparts. This suggests that cultural factors play a significant role in shaping individuals' perceptions and responses to living density, influencing their overall well-being in residential environments.

In the next lecture we will elaborate about the environmental changes and rural areas.

Lesson 37

Environmental Changes in Rural Areas

Rural Areas and Their Characteristics

Rural areas are characterized by their strong connection to community and nature. However, it's argued that no truly untouched natural places exist anymore; human influence has shaped all rural landscapes.

- **Rurality and Nature:** Historically, there's been a close link between nature and society, with rural areas often associated with a purer and nobler form of living compared to urban spaces.

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- **Romanticized Association:** Rural landscapes are often seen as natural, and rural activities are perceived as working harmoniously with nature. Rural inhabitants are often seen as more attuned to and respectful of nature.

Factors Contributing to Environmental Changes

Following factors contribute to the environmental changes in rural areas:

- **Agricultural Practices:** The use of pesticides, chemical fertilizers, and the removal of hedgerows have led to increased yields but also contributed to habitat destruction and reduced plant diversity.
- **Urbanization:** The expansion of urban areas into rural spaces has led to loss of open space, increased pollution, demand for infrastructure like roads and drainage systems, and light and noise pollution.
- **Forestry and Primary Production:** Deforestation, afforestation of open areas with nonnative species, mining and quarrying, and flooding for reservoirs have all impacted rural environments.

Degradation of Natural Environment

The causes of the degradation of the natural environment are as follows:

- **Modern Agriculture:** Intensive farming practices have led to declines in plant and animal species due to habitat destruction and chemical use.
- **Tourism Impact:** Tourists attracted to rural landscapes have contributed to erosion, pollution, and construction-related land loss.
- **Global Environmental Change:** Rural areas also face challenges from global warming and other environmental shifts.

Approaches to Dealing with Rural Environmental Change

The two main approaches to deal with the rural environmental change are given below:

- **Utilitarian Perspective:** Some argue that nature can adapt to a certain level of change, and not all environmental alterations are problematic.
- **Natura-Ruralistic Perspective:** Others believe that environmental changes have already caused irreparable damage, and urgent action is needed to prevent further harm.

Course of Action and Challenges

Following course of action to be taken in order to deal with the effect of environmental changes occurring in rural areas are:

- To protect wildlife and habitats, farming practices may need extensive regulation.
- Utilizing alternate energy sources like wind power can significantly impact rural environments.

Despite various measures and programs, finding the right response to rural environmental changes remains a significant question influenced by political and economic factors.

Lesson 38

The Built Environment and Human Adjustment

In environmental psychology, the built environment plays a crucial role in shaping human behavior and adjustment. The design of institutions such as hospitals, prisons, and educational environments significantly impacts individuals' experiences and interactions within these spaces. Let's delve into how institutional design influences human behavior and adaptation.

Understanding Queuing Theory: The Never-Ending Wait

Lines or queues are a ubiquitous aspect of daily life, from grocery stores to fast-food restaurants. The experience of waiting in line can be frustrating and time-consuming, leading to questions about the design of these spaces.

Vile's Laws of Advanced Linesmanship humorously capture the nuances of queuing behavior, highlighting challenges like line length, unexpected delays, and social interactions in queues. Here are funny rules that explain how people behave while waiting in lines:

- If you are running for a short line, it suddenly becomes a long line.
- If you are waiting in a long line, people behind you are shunted to a new short line.
- If you step out of a short line for a second, it becomes a long line.
- If you are in a short line, the people ahead of you let in their friends and relatives and make it a long line.
- A short line outside a building becomes a long line.
- If you stand in one place long enough, you make a line.

These laws humorously reflect the frustrations and unpredictability often associated with waiting in queues or lines. They highlight common experiences and observations people make when navigating through lines in various settings.

The Role of Good Architecture in Public Spaces

Architectural design shapes our experiences in public spaces such as supermarkets, airports, and hospitals. Good architecture minimizes human discomfort and maximizes functionality, yet achieving this balance requires understanding users' needs and preferences. C.M. Deasey's insights underscore the importance of aligning design with human nature and demands for change in public attitudes toward architecture.

General Hospitals

The hospital environment encompasses a variety of people, including patients with diverse needs and medical staff with specialized roles. However, conflicts can arise between the needs of patients and staff, highlighting the importance of balancing efficiency with patient well-being. For instance, during surgery, patients require a specific atmosphere, while staff members are under stress and may need different environmental conditions.

A Mental Hospital – Example of Design

Robert Sommer's observations in a mental hospital shed light on the impact of design on patient well-being. The installation of new amenities without consulting patients led

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to a detrimental arrangement that hindered social interaction. This case underscores the need for inclusive design practices that consider user input and promote positive experiences.

Key Factors in Hospital Design

Examining hospital design reveals critical factors influencing human behavior, such as proximity, privacy, and control. Design choices like the location of nurses' stations, room layouts, and environmental controls significantly affect staff efficiency and patient satisfaction. Understanding these dynamics can lead to improvements in hospital design that enhance both functionality and user experiences.

Hospital design should aim to maintain behaviors, encourage healthy behaviors for rehabilitation, and provide freedom for individuals to operate and realize their potential. The environment must compensate for individual failures and enhance performance levels where possible, striking a balance between efficiency and patient care.

Hospital Design and Its Effects

As environmental psychologists, studying institutional design allows us to understand how physical spaces shape behavior, emotions, and interactions. By analyzing real-world examples and research findings, we gain insights into creating environments that promote well-being, productivity, and positive social dynamics. Institutional design serves as a lens through which we examine the intricate relationship between humans and their built surroundings.

Effective hospital design directly influences user behavior and experiences. Observing how the physical layout of a hospital affects daily activities can reveal opportunities for improvement. For instance, assessing the accessibility of facilities, signage clarity, and ease of navigation can lead to design suggestions that enhance user experience and efficiency.

Conclusion: Bridging Design and Human Experience

In conclusion, the built environment profoundly influences human behavior and adjustment in institutional settings. Through a multidisciplinary approach that considers user needs, feedback, and empirical research, environmental psychologists contribute to designing spaces that optimize human experiences and outcomes. By studying institutional design, we uncover valuable insights into fostering environments that support well-being, functionality, and meaningful interactions.

In the next lecture we will talk about the prisons.

Lesson 39

Prisons

In the realm of correctional facilities, there exists a complex interplay of conflicting goals and varied philosophies regarding the purpose and design of prisons. This handout delves into the intricacies of imprisonment, exploring different models,

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architectural ideologies, and their impact on inmates' experiences.

Conflicting Goals of Imprisonment

Prisons face conflicting goals and perspectives, resulting in disparities in sentencing and rehabilitation opportunities.

Diverse Perspectives: Various stakeholders, including corrections employees, judges, legislators, and inmates, hold conflicting ideas about the goals of prisons.

Sentencing Disparity: This diversity of perspectives leads to significant differences in sentencing, parole criteria, and rehabilitation opportunities.

Philosophical Models of Imprisonment

Let's explore different philosophical models of imprisonment and their impact on prison design. **Sommer's Models:** Robert Sommer delineates several models of imprisonment, such as deterrence, rehabilitation, retribution, and integration, each with its unique implications for prison design. Each model represents unique challenges in prison design and management, influenced by its underlying philosophy. These conflicting viewpoints contribute to sentencing disparities, parole variations, divergent vocational and psychological support, and discrepancies in actual time served for similar offenses, termed as a "model muddle" by Siegler and Osmond (1974).

Architectural Adaptations: The design of prisons varies significantly based on the model being followed. For instance, an integration-focused prison may include vocational workshops and educational facilities, while a deterrence-oriented one may prioritize security features.

Life inside Prisons

Exploring the dynamics of life within prison walls unveils intriguing analogies between prison environments and hospital settings, highlighting the contrasting philosophies and approaches to design.

Hospital vs. Prison Analogies: The physical design of prisons symbolizes containment and control, akin to hospital settings but with stricter regulations and limited personal space. **Historical Architectural Alignment:** Traditional prison architecture aligns with deterrence and retribution philosophies, resulting in impersonal and restrictive environments for inmates. **Modern Approaches:** Contemporary perspectives advocate for prison designs aligned with reeducation, reform, and integration philosophies. Examples like New Jersey's Leesburg prison showcase attempts to create more humane and rehabilitative environments through innovative architectural features.

Empirical Insights

The empirical findings presented in the following studies are as follows:

- Gilbert's study in 1972 emphasized that limiting inmates' movement within the prison environment is more crucial than the actual size of their cells.
- Sommer's research in 1972 highlighted that factors such as crowding, lack of privacy, and sensory deprivation contribute significantly to the challenges of confinement and restriction faced by prisoners.
- Luxenberg's analysis in 1977 indicated that modern prisons have made efforts to mitigate the negative effects of designing solely for maximum security, achieving varying levels of success in this endeavor.

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- Paul Paulus and colleagues found that inmates in single or double occupancy cells perceive less crowding compared to those in cells with larger numbers. Consequently, prisoners in denser cells report more health issues and spend more time in the infirmary.
- Schaeffer, Baum, Paulus, and Gaes discovered in 1988 that partitioning cubicles in prison cells led to a reduction in urinary catecholamine levels, which is an indicator of stress among inmates.
- Wener and Keys' research in 1988 revealed that increased population density within prison cells is associated with higher rates of perceived crowding, more sick calls, and increased isolated behavior among inmates.
- Eckland's study in 1986 suggested that violence in prisons, often linked with high density, is less about cognitive confusion and tension and more about issues of control within the prison environment.

However, challenges persist in fostering a cooperative and rehabilitative prison environment, emphasizing the ongoing need for research and progressive design strategies.

Case Study

Charles in City Prison: Analyzing individual experiences like Charles' incarceration sheds light on how prison design may influence inmate behavior and rehabilitation outcomes. Charles, a 34-year-old individual, was arrested for driving 41mph in a 25 mph zone. Despite having a clean record regarding alcohol and drugs, he was found with a small quantity of cocaine in his car. As a result, he faced additional charges for possessing an illicit drug. Charles was sentenced to two years in city prison.

During his time in city prison, Charles was housed in a single cell designed for two people. He encountered a series of cellmates who had received sentences ranging from three days to life imprisonment. Throughout his incarceration, Charles learned about new drug dealers and methods of drug use that he had not been aware of before. This experience raises questions about the potential influence of prison design and environment on inmates' exposure to criminal behaviors and their subsequent behavior post-release.

Conclusion

Reflecting on questions about the relationship between design, punishment philosophy, and inmate behavior encourages critical thinking and exploration of reformative approaches. Prison design is not merely about physical structures but also about embodying philosophical ideals and fostering environments conducive to rehabilitation and integration. Understanding these complexities is essential for informed discussions in environmental psychology and criminal justice studies.

In the next lecture we will talk about the education environments.

Lesson 40

Education Environments

When discussing the educational environment, the classroom holds significance within

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that context. This lesson highlights how the physical design and layout of classrooms influence various aspects of the educational experience, such as class participation, student behavior, and academic performance.

The built environment refers to the physical structures and spaces that humans create for living, working, and learning. In the context of education, the built environment encompasses classrooms, lecture halls, libraries, dormitories, and other educational facilities. These physical spaces play a crucial role in shaping the overall educational experience and can significantly impact students' learning outcomes.

Traditional Classroom Arrangements

The traditional classroom setup, characterized by rows of students facing an instructor, reflects the modal arrangement in educational environments. This layout, with its genesis in utilizing outside light for illumination, serves several purposes such as ease of surveillance, attendance tracking, and maintaining control by the instructor.

Seating Arrangement

Seating positions within the classroom can affect class participation, attentiveness, and academic achievement.

- **Impact on Student Behavior:** Studies by various researchers have shown the influence of seating arrangements on student behavior. For instance, Sommer's research indicates that students in the front row tend to participate more, while Koneya's findings suggest that low verbalizers may participate less regardless of their seating position.
- **Attention and Academic Performance:** Further studies by Schwebel, Cherlin, and others highlight that students in front-row seats are more attentive and often achieve higher grades, even when seating positions are assigned rather than chosen by students.
- **Designing for Interaction:** In smaller classrooms or seminar-type settings, seating positions directly influence participation levels. For example, students seated opposite the instructor tend to participate the most, emphasizing the importance of seating proximity in facilitating interaction.

Color and Environmental Factors

Research by Kuller and Wilson suggests that colors can influence psychological processes, mood, and attitudes, thereby indirectly impacting students' learning experiences. Blue influences feelings of security. Red is stimulating and leads to excitement. Colors influence blood pressure and respiration rate.

Class room behavior, performance and learning will be influenced by such environmental conditions as:

- **Temperature:** Temperature plays a crucial role in creating a comfortable and conducive learning environment. Extreme temperatures, whether too hot or too cold, can distract students and affect their focus and concentration. Maintaining an optimal temperature in the classroom helps students stay attentive, engaged, and more receptive to learning.
- **Noise:** Similarly, noise levels in the classroom can have a profound impact on student performance. Excessive noise, such as chatter from neighboring

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classrooms, outdoor sounds, or mechanical noises, can disrupt concentration and hinder effective communication between students and teachers. On the other hand, a quiet and controlled acoustic environment allows for better concentration, improved comprehension, and enhanced classroom interactions, leading to more productive learning experiences.

- **Crowding:** High levels of crowding in educational environments have a profound impact on students, affecting behavior, performance, and learning outcomes. This crowding often results in discomfort, limited personal space, and heightened stress levels among students, which can disrupt their focus and engagement in academic activities. Furthermore, it hampers social interactions and collaborative efforts, thereby impairing peer learning and teamwork.
- **Light (illumination):** As illumination increase, visual activity increases. With more light we are able to recognize and detect smaller visual details. Effects of changes in illumination are more pronounced on difficult tasks than on easy tasks. Greater illumination allows for more accurate and quick discrimination. At very high levels of illumination performance decrements are likely because additional light can act to suppress some information cues such as visual gradients.

Open Space Classrooms

Exploring innovative classroom designs, open-space classrooms offer flexibility and increased interaction among students and teachers. However, studies also highlight challenges such as increased noise levels and visual distractions, emphasizing the need for careful design considerations.

Optimizing Learning Environments

Effective classroom design involves balancing factors like illumination, noise control, and seating arrangements to create conducive learning environments. By addressing these design elements, educational settings can maximize learning potential and student engagement.

Considerations for Dormitory Design

Beyond classrooms, dormitory design also plays a crucial role in students' educational experiences. Research indicates that factors like social density, noise levels, and personal control within dormitories can significantly impact students' well-being and academic performance.

Conclusion

In summary, the physical design of educational environments, including classrooms and dormitories, can significantly influence learning outcomes, student behavior, and overall academic success. By understanding and addressing environmental factors, educators and designers can create spaces that enhance the learning experience and promote student.

Lesson 41

Institutional Design Reconsidered & Queuing Theory

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In this lecture, we delve into the critical aspects of institutional design within the realm of environmental psychology. We explore the complexities involved in designing spaces that cater to the diverse needs and behaviors of their users, emphasizing the importance of considering human welfare in the built environment.

Understanding Institutional Design

- **Complexity of Design** o Highlighting the limitations of oversimplified design paradigms.
 - o Emphasizing the need for comprehensive data analysis to inform design decisions.
- **Interaction of Physical and User Characteristics** o Discussing how the performance of a built environment is influenced by its physical attributes and user requirements.
 - o Acknowledging the varied needs of multiple users in designed environments.
- **Evaluation Beyond Prototypes** o Stressing the importance of evaluating designs based on actual user experiences rather than theoretical models.

Challenges in Design Process

- **Technical Focus of Architects and Designers** o Exploring the primary concerns of architects and designers in terms of technical and structural aspects.
 - o Recognizing the potential gaps in understanding user cultural dynamics and ecological relationships.
- **Cultural and Ecological Considerations** o Discussing the significance of factors like privacy, territoriality, and adaptation in design.
 - o Addressing cross-cultural nuances that impact design and social change.

Steps in Design Process

- **Preparation and Problem Statement:** Defining the problem statement and design objectives clearly.
- **Data Gathering and Analysis:** Emphasizing the importance of gathering relevant data for informed decision-making.
- **Outcome Evaluation and Creativity:** Defining desirable outcomes and exploring cost-effective design solutions.
- **Selection and Implementation:** Decision-making on design alternatives and transforming plans into physical structures.
- **Reevaluation and Adaptation:** Assessing design performance and refining structures based on user feedback.

Considerations for Design Criteria

- **Performance Standards:** Discussing criteria such as profit, quality, performance, competition, compatibility, flexibility, elegance, safety, and time in setting design standards.
- **Human Factors and Physical Limitations:** Exploring design prescriptions based on human physical and anatomical characteristics.

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Queuing Theory and Its Applications

Queuing theory, focusing on waiting dynamics, is vital for efficient design in various contexts, from retail settings to transportation systems. Understanding traffic intensity and optimizing service processes can significantly enhance user experiences.

- **Queue and Traffic Intensity:** Explaining the concept of queue and traffic intensity in service environments.

Example: Illustrating queue dynamics with an example of customers at a grocery store. This example demonstrates how queuing theory can be applied to analyze and optimize customer service in a grocery store setting. Store managers can use this data to anticipate customer arrival intervals, optimize service efficiency, and make informed decisions such as opening more registers when required.

Recommendations for Store Managers is to use strategies for anticipating customer arrival intervals and optimizing service efficiency.

- **Other Uses of Queuing Theory:** Queuing theory has utility in any situation where all potential users of a facility cannot utilize the facility simultaneously. It goes beyond service industries and can be applied to various scenarios to improve system efficiency and user experiences.

o**Accessible Environments for Persons with Disabilities:** Revisiting discussions on creating inclusive environments.

o**Energy Use and Behavioral Change:** Analyzing energy demands and the need for behavioral shifts. Discussing grassroots efforts and community-based initiatives for energy conservation.

- **The Future Challenges:** Energy use and abuse, environmental costs of energy and material usage, strategies for saving energy, alternate energy resources, behavioral solutions to environmental problems, environmental attitudes and behaviors, integration, and looking to the future are some of the challenges that lie ahead. Understanding these challenges and working towards sustainable solutions is crucial for a better future.

Lesson 42

Energy Use in Homes and Commercial Buildings

Determinants of Energy Use at Homes

Heating and cooling systems play a crucial role in determining energy consumption in homes. For instance, using a natural gas furnace for heating is more energy-efficient than relying on electrical baseboard heating. This efficiency stems from reduced energy loss during the conversion of primary energy sources like coal, fuel, oil, or natural gas into electricity. Additionally, lifestyle factors significantly influence energy use at homes. The number and types of electrical appliances, such as color TVs versus black and white TVs, frost-free refrigerators versus conventional ones, air conditioners versus window fans, and incandescent lights versus fluorescent lights, all impact energy consumption levels.

Savings for Consumers – An Example

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Considerable economic and environmental benefits arise from choosing energy-efficient appliances and practices. For example, if 6.6 million refrigerators/freezers sold in 1970 were energy-efficient, it would lead to substantial savings in electricity and dollars. This would not only benefit consumers economically but also result in significant reductions in environmental pollution, including sulfur dioxide, nitrogen oxides, and reduced land use for power plants and transmission lines.

Home Construction and Energy Use

The construction of homes is a critical factor in determining energy consumption. Key considerations include wise use of insulation, effective shading in summer, and utilization of sunlight in winter. Implementing these strategies can result in substantial energy savings without compromising comfort. For instance, well-insulated homes with strategic shading can retain heat during winter and keep interiors cool during summer, leading to reduced energy needs for heating and cooling.

Energy Use in Commercial Buildings

Commercial buildings often exhibit excessive energy consumption due to design choices such as extensive glass usage and tall structures. For example, buildings like the Sears Tower in Chicago consume more electricity than entire towns. Implementing energy-efficient design principles and modifying appliance usage can lead to significant energy savings in commercial buildings.

Environmental Costs of Energy and Materials Usage

The use of energy and materials has far-reaching environmental impacts throughout extraction, processing, and usage stages. Oil drilling activities, for instance, can result in blow-outs, spills, and refinery pollution, causing harm to marine life and ecosystems. Addressing these environmental costs requires adopting sustainable practices and reducing reliance on nonrenewable resources.

Strategies for Saving Energy and Promoting Sustainability

Efforts to save energy and promote environmental sustainability include using and producing less, extending the lifespan of products through reuse and recycling, and choosing energy-efficient materials and appliances. Embracing renewable energy sources like solar energy and implementing energy-saving measures in construction and daily practices are crucial steps towards a more sustainable energy future. However, challenges such as initial investment costs and the availability of existing energy sources need to be addressed to facilitate widespread adoption of sustainable energy practices.

Lesson 43

Conservation as Energy Source

Three Types of Conservation

- **Curtailment:** Curtailing energy use involves reducing the overall consumption of energy resources. This can be achieved through conscious efforts to use energy-efficient appliances, adopting energy-saving practices, and minimizing wastage.
- **Overhaul:** Overhauling refers to making dramatic changes in lifestyle and societal norms to reduce energy consumption. An extreme example of this would be implementing laws against suburbanization, which can significantly impact energy demands.
- **Adjustment:** Adjustment involves making specific changes to enhance energy efficiency. This includes insulating homes, improving the efficiency of automobiles and home appliances, and capturing waste heat for productive use.

Imagine Your Life 50 Years from Now

Visualize a future where energy conservation and efficiency are ingrained in everyday life, leading to sustainable and eco-friendly practices across all sectors.

Behavioral Solutions to Environmental Problems

- **Tragedy of Commons:** The concept of the tragedy of the commons highlights the conflict between immediate self-interest and future public interest. It underscores the need for collective action and responsible environmental stewardship.
- **Education to Preserve Environment:** Educational initiatives play a crucial role in promoting environmental preservation. However, simply telling people what to do may not be effective. Instead, efforts should focus on changing attitudes and behaviors gradually.
- **Applied Behavior Analysis and Intervention:** Applying principles of behavior modification can lead to positive changes in environmentally relevant behaviors. Strategies include:
 - o **Baseline Period:** Observing naturally occurring behaviors related to energy use.
 - o **Treatment Period:** Introducing interventions like rewards for desired behaviors and punishments for non-target behaviors.
 - o **Follow-Up Period:** Continuing observations post-intervention to assess effectiveness.

Intervention Strategies

- **Antecedent Strategies:** Using prompts and cues, such as written messages encouraging eco-friendly actions, can trigger desired behavioral responses. Specific prompts near the intended behavior location and polite messaging are more effective.
Examples of prompts:
"Help keep your pool clean."
"Do not litter."
"Please avoid littering."
- **Consequent Strategies:** These strategies involve presenting or removing

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pleasant/unpleasant consequences following behaviors to reinforce or discourage them.

Types of consequences include:

- o **Reinforcement:** Increasing the frequency of desired behaviors.
- o **Punishment:** Decreasing the frequency of undesirable behaviors.

Types of reinforcement:

- o **Positive Reinforcement:** Offering pleasant consequences for desired behaviors.
- o **Positive Punishment:** Imposing unpleasant consequences for undesirable behaviors.
- o **Negative Reinforcement:** Removing negative consequences for desired behaviors.
- o **Negative Punishment:** Removing pleasant events for undesirable behaviors.

Integration and Look to the Future

Our collective behaviors significantly impact environmental quality. By adopting energy-efficient practices and promoting sustainability, we contribute positively to long-term public interest and mitigate environmental degradation caused by excessive energy consumption.

Lecture 44

Understanding Urban Sprawl: Challenges and Solutions

Further Plans – The Cities

- **Urban Sprawl:** Urban sprawl refers to the expansion of low-density residential areas, shopping centers, and industrial facilities connected by multilane highways, leading to unplanned development and environmental challenges.
- **Origins of Urban Sprawl:** Historically, cities were designed for walking, bicycles, and public transport, with compact neighborhoods offering easy access to amenities. However, with technological advancements and increased private transportation, cities expanded outward.
- **The Dream of Suburbs:** The desire for personal space and the convenience of private transportation led to a housing boom in suburbs. This shift from city centers to suburban areas resulted in haphazard development and increased dependence on cars.

Urban Blight

- **Challenges Faced:** Industrialization brought challenges like poor housing, pollution, and congestion in cities, prompting people to seek housing outside urban centers.
- **Consequences:** The rapid growth of housing colonies without proper planning led to a maze of developments, straining local governments to provide essential services like schools, water, and roads.
- **Vicious Cycle of Development:** Efforts to reduce congestion with new highways only fueled further sprawl, creating a cycle of increased commuting distances and traffic congestion.

Moving Towards Sustainable Communities

- **Smart Growth Initiatives:** To address these challenges, smart growth initiatives promote compact, walkable communities with mixed land use, reducing environmental impacts and improving quality of life.
- **Environmental Impacts of Urban Sprawl:** Urban sprawl leads to energy depletion, air and water pollution, loss of agricultural land, and fragmentation of wildlife habitats, impacting ecosystems and human health.

Towards the Common Good

- **Quality of Life vs. Environmental Concerns:** While sprawl offers benefits like larger homes and better services, it also increases vehicle ownership, fatal accidents, and health issues like obesity and high blood pressure.
- **Common Good Perspective:** Environmental costs are often overlooked in favor of personal benefits, highlighting the need for collective responsibility and

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sustainable choices for common good.

- **Reining in Urban Sprawl: Smart Growth Solutions:** By promoting compact development, public transport, and green spaces, communities can mitigate the negative impacts of sprawl and create livable, eco-friendly environments.

In conclusion, understanding the complexities of urban sprawl is crucial for creating sustainable cities that balance personal preferences with environmental conservation for the common good.

Lesson 45

Urban Blight

In the developed world the other end of ex-urban migration is the city from which people are moving. A completely different set of factors is responsible for urban blight in developing countries. In this case, people are moving *to* the cities at a rate that far exceeds the capacity of cities to assimilate them. The result is urban slums that surround virtually every city in the poorer developing countries.

The economies of rural areas, often based only on subsistence farming simply do not provide the jobs needed by a growing population. So, people move to the cities, where at least they have the hope for employment.

City Housing

City housing is overwhelmed by the influx of migrants who could not afford the rents even if housing was available this leads to the formation of slums

Slums Surrounding the Cities

Vast slums surrounding the cities are a great challenge to the institutional structure of developing countries. Because slums are unauthorized areas, cities rarely provide them with electricity, water, sanitation, and other social amenities. Yet the slums often represent essential workforce to the city taking low-paying jobs that keep the city and its inhabitants functioning. People live in the fear of bulldozers coming in any time and leveling down the shanty town. A great need in such neighborhood is home security. People in the slums also need more jobs. Greatest need for people living in these areas is government representation. Their voices are seldom heard because they are among the poorest and most powerless in society.

What Makes Cities Livable

A sustainable future will depend upon both reining the urban sprawl and revitalizing the cities. The urban blight in the cities in developing countries requires deliberate policies to address the social needs of people flocking to the shanty towns. The only possible way to sustain the global population is by having viable, resource-efficient cities, leaving the countryside for agriculture and the natural ecosystem.

Viable

Viable means livable. No one wants to and should be required to live in the conditions that have come to typify the urban blight. Livability is a general concept based on people's response to the question "Do you like living here, or would you rather live somewhere else?"

Livable Cities around the World

The common denominator of livable cities in the world is: Maintaining high population density. Preserving heterogeneity of residences, businesses, stores, and shops keeping layouts on human dimension so that people can meet, visit or conduct business incidentally over coffee at side walk café or stroll through an open area.

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People VS Automobiles

In livable cities the space is designed for and devoted to people. In contrast, development of the last fifty years has concentrated on accommodating automobiles and traffic. Two thirds of the land in the cities that have grown up in the era of automobiles is devoted to moving, parking or servicing the cars.

A Matter of Design

The world's most livable cities are not those with perfect auto access between all points. Instead they are the cities that have taken measures to reduce outward sprawl, diminish automobile traffic, and improve access by foot and bicycle in conjunction with mass transit.

Examples of Cities around the World

Geneva, Switzerland prohibits automobile parking at workplaces in the city's center forcing customers to use excellent public transport system. Copenhagen bans all on-street parking in the downtown core. Paris has removed 200,000 parking places in the downtown area. Curitiba, Brazil with a population of 1.6 million is cited to be one of the most livable city in the entire Latin America because they have guided development around the idea of mass transit system rather than cars.

Livable = Sustainable

Livability of a city leads to its sustainability. Reduction of auto traffic and greater reliance on foot and public transportation reduce energy consumption and pollution. Urban heterogeneity can facilitate the recycling of materials. Housing can be retrofitted with passive solar space heating and heating for hot water. Landscaping can provide cooling. Vacant or cleared areas can be converted into garden plots. Rooftop hydroponic gardens can be made popular. Such gardens may not make the cities agriculturally self-sufficient but they add to urban livability. By making the urban areas more appealing and economically viable we not only improve the lives of city residents but also spare the surrounding areas. Parks, wilderness, and farms will not be replaced by exurbs and shopping malls but rather will be saved for future generations.

Moving Towards Sustainable Communities

A joint facility of the UN Environment Program and the UN Human Settlement Program, the Sustainable Cities Program (SCP) is designed to foster the planning and management of needed to move cities in developing countries towards sustainability.

Sustainable City

Sustainable Cities Program (SCP) defines the sustainable city as a city in which achievements in social, economic, and physical development are made to last. Cities are the focus of the program because they are absorbing two-thirds of population growth in developing countries and in the process, we are experiencing serious environmental degradation in and around the growing urban centers.

Bottom-up Approach

The bottom-up approach calls for the involvement of all people at all economic and social

Be kind To Yourself.

levels and reconciling their interests when conflicts are evident Common principles seen in sustainable development theory are principles such as Social equity, Economic efficiency, and Environmental planning and management.

SCP Cities

To date, there are twenty SCP cities some of them are: Madras (India), Dar es Salam (Tanzania), Accra (Ghana), and Shenyang (China), Concepcion (Chile).

Towards the Common Good

How must we live on our planet? There are three strategic themes to answer the question: Sustainability, Stewardship, and Sound Science.